Comparing seasonal and latitudinal patterns in thermal adaptation

2026-10-23

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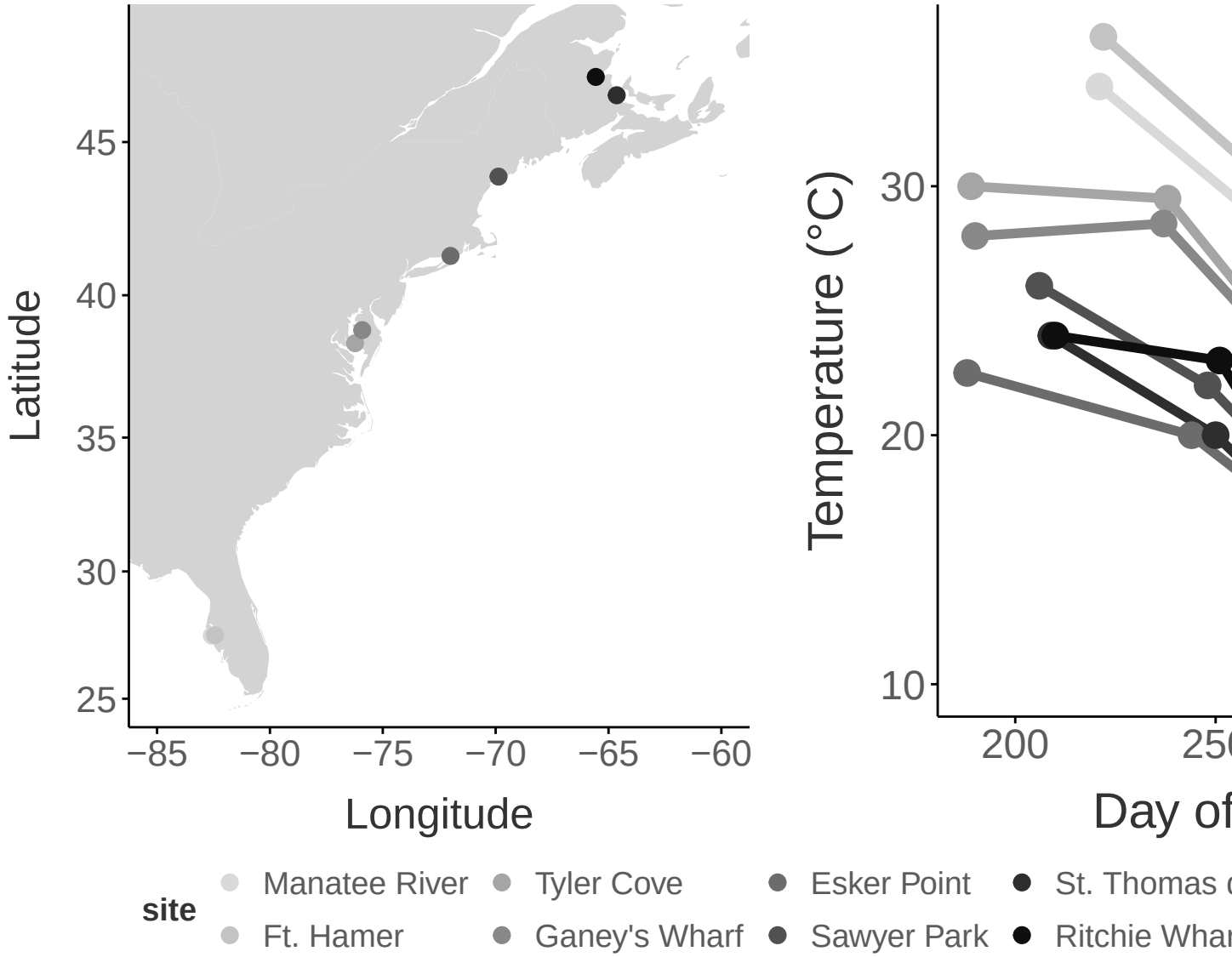
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Main Message

By assaying thermal limits several times per year in natural populations distributed across a large latitudinal gradient, we highlight the importance of intra-specific variation for predicting vulnerability to climate change. Indeed, across the entire spatiotemporal framework examined here, we observed an ~12°C range in thermal limits. Both spatial and temporal variations in thermal limits are essential for robust predictions. Also highlighted in this study are the major differences in thermal limits between the various cryptic lineages of *Acartia tonsa*. The broad (and sometimes overlapping) distrbutions of these clades provide a unique opportunity to compare replicate spatiotemporal patterns in adaptation against a relatively similar genetic background.

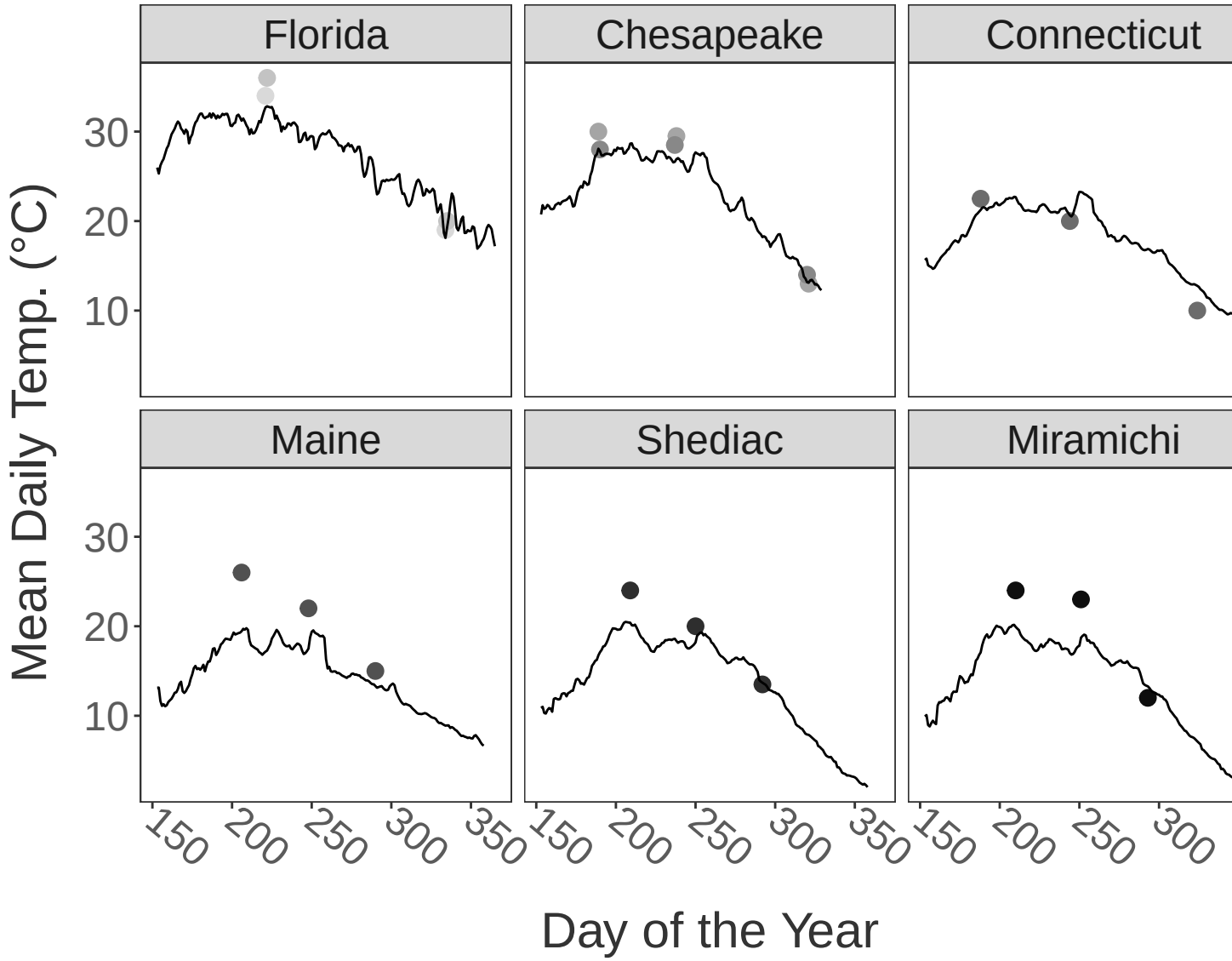
Site Characteristics

Copepods were collected by surface tow from sites across the Western Atlantic at several times throughout the year. The sites are shown below along with surface temperatures and salinity were measured at the time of collection. Across the entire set of collections, temperature ranged from 10°C to 36°C.



Collections aimed to obtain copepods near the onset of peak temperatures, after peak temperatures, and then after temperatures cooled during the Fall. To put the collections into this larger seasonal context, we examined continuous temperature records for each site. Continuous data is not available for all sites, however, so here we've pieced together daily temperature values from either local temperature sensors (sites in Florida and the Chesapeake) and high resolution satellite temperature data (Connecticut, Maine, and the Canadian sites). This satellite data comes from the NOAA 1/4° Daily Optimum Interpolation Sea Surface Temperature (OISST).

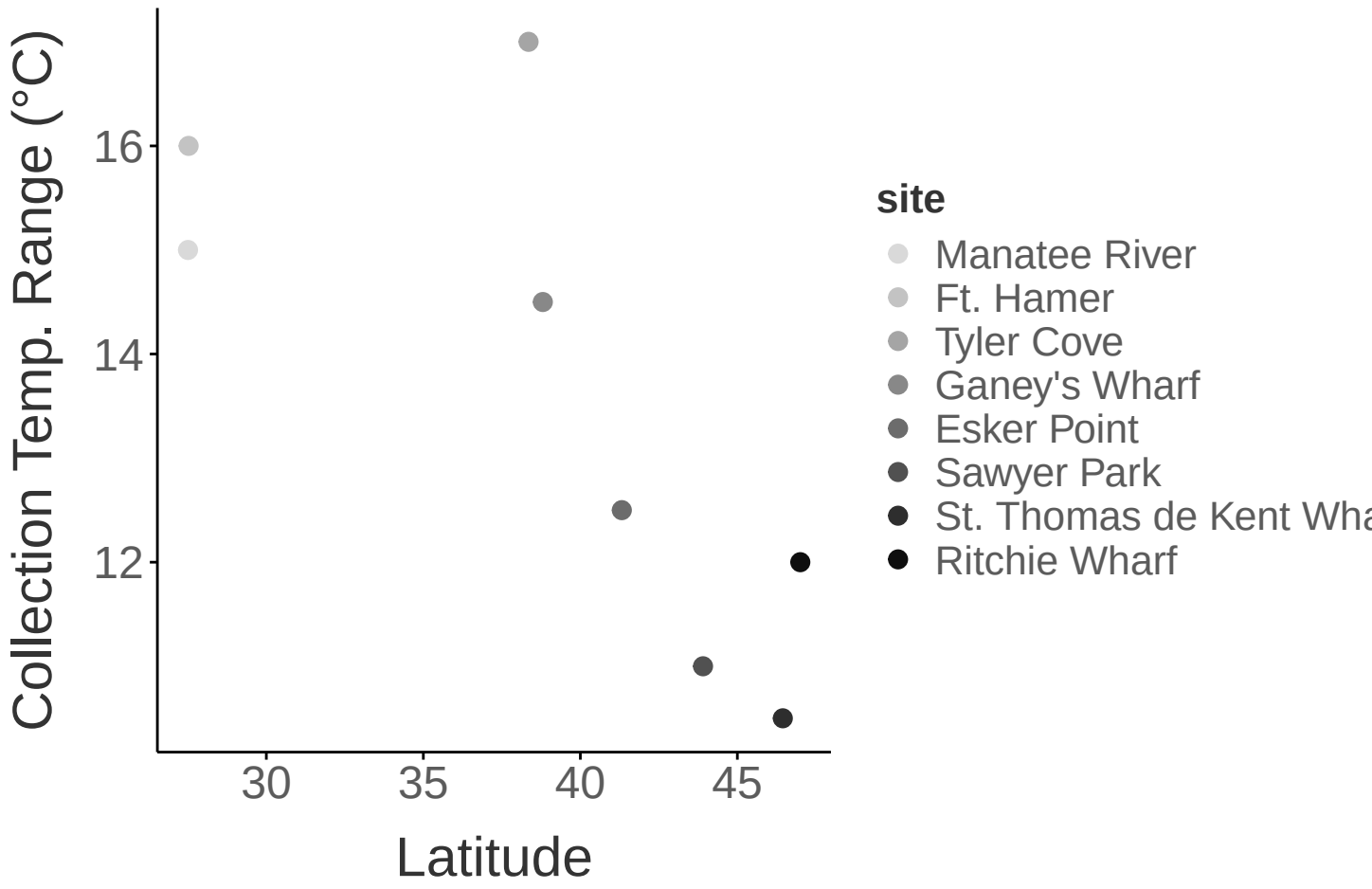
These temperature profiles are shown below, with the temperatures measured during the time of collection included for comparison While collection temperatures in Maine did not always match the recorded daily averages, across all sites the temperature records do give a general sense of the timing of seasonal maxima (and the proximity of collections to the seasonal maximum temperature). In general, the first sample from each site fell just after the site reached the warmest period. The exception to that pattern is in Florida, where collections occurred after an extended period of high temperatures.



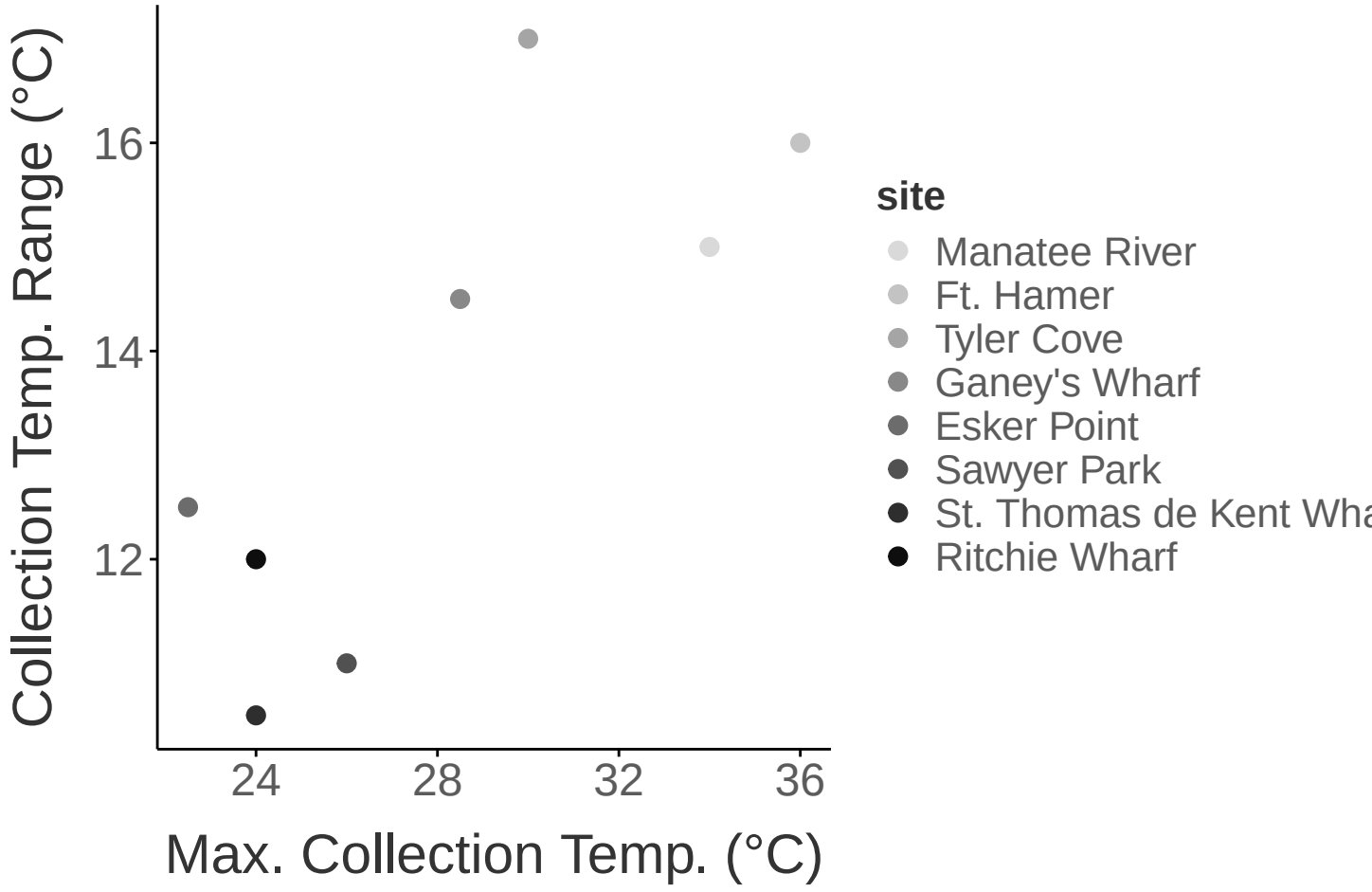
Exact locations for the sites are provided here:

Site	Region	Lat	Long
Key Largo	Florida	25.28391	-80.33014
Manatee River	Florida	27.50561	-82.57277
Ft. Hamer	Florida	27.52488	-82.43101
Tyler Cove	Maryland	38.35083	-76.22902
Ganey's Wharf	Maryland	38.80555	-75.90906
Esker Point	Connecticut	41.32081	-72.00166
Sawyer Park	Maine	43.90698	-69.87179
St. Thomas de Kent Wharf	New Brunswick	46.44761	-64.63692
Ritchie Wharf	New Brunswick	47.00481	-65.56291

The latitudinal gradient covers a wide range of seasonality. Shown below is the temperature range. While based on collection temperatures, and therefore an underestimate of the total seasonal range of temperatures, these patterns are representative of the expected latitudinal gradient in seasonality.



There was a general increase in the collection temperature range at warmer sites. This provides an opportunity to disentangle some of the traditionally proposed hypotheses about thermal adaptation (macrophysiological gradients often include stable and warm environments contrasted with cooler, more variable environment).



Phenotypic Measurements

Critical Thermal Limits

A total of 427 individuals were examined. Critical thermal limits and body size measurements were made before individuals were preserved in ethanol. We excluded data for individuals that were either later classified as males or juveniles, that we were unable to assign to a clade, or that were identified as individuals from a closely related species (*Acartia hudsonica*). With these individuals excluded, the full data set contains 397 adult females with CTmax and body size measurements, and clade identifications.

Critical thermal maxima (CTmax) was measured using a custom setup. The method uses a standard dynamic ramping assay to determine the maximum temperature individuals could sustain normal functioning. This differs from lethal temperatures, and indeed, all individuals recovered following the assay.

Individuals were rested for one hour after collection before the assay. During the assay, copepods were held in artificial seawater, composed of bottled spring water and Instant Ocean salt mix adjusted to match collection salinities. During the assay, several 'control' individuals were maintained in this solution at ambient temperatures without the temperature ramp to ensure that there was no background changes in behavior or mortality. When sorting individuals from the plankton tow contents, they were held in a 50:50 mix of 60 um filtered water from the collection site and artificial seawater as an additional acclimation step.

Sample sizes varied slightly across experiments, but generally ~20 individuals were measured per season at each site. The major exceptions were the early samples from the Florida sites and the late sample from Sawyer Park (Maine). Only two sets of samples (peak and late) were collected from Fort Hamer and Manatee River. The late season collection from Sawyer Park occurred after *Acartia tonsa* abundance decreased, and samples were dominated by *Acartia hudsonica*. Thermal limits were measured for all *A. tonsa* individuals

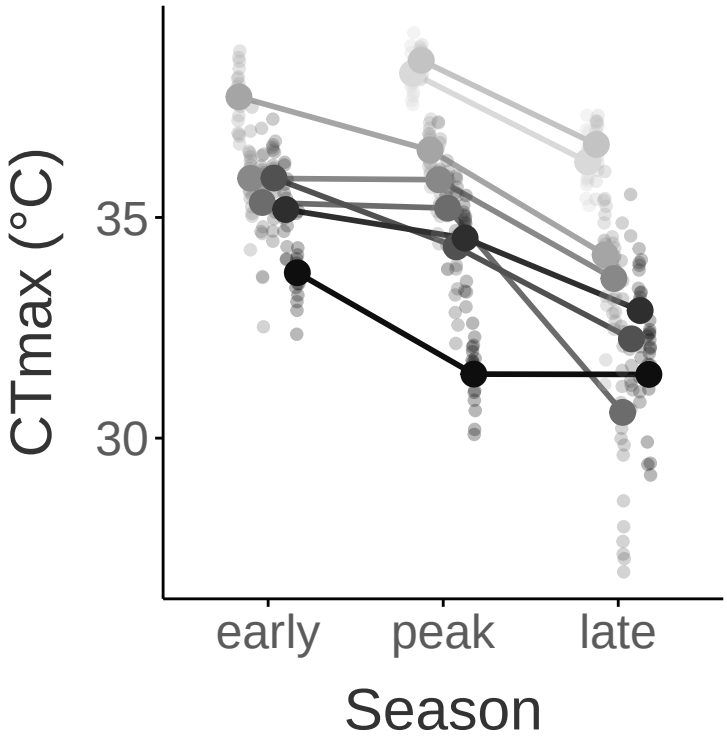
recovered. No samples were collected from Key Largo for this project, as *Acartia tonsa* wasn't present in the water during the peak season, likely due to a recent extreme heat event.

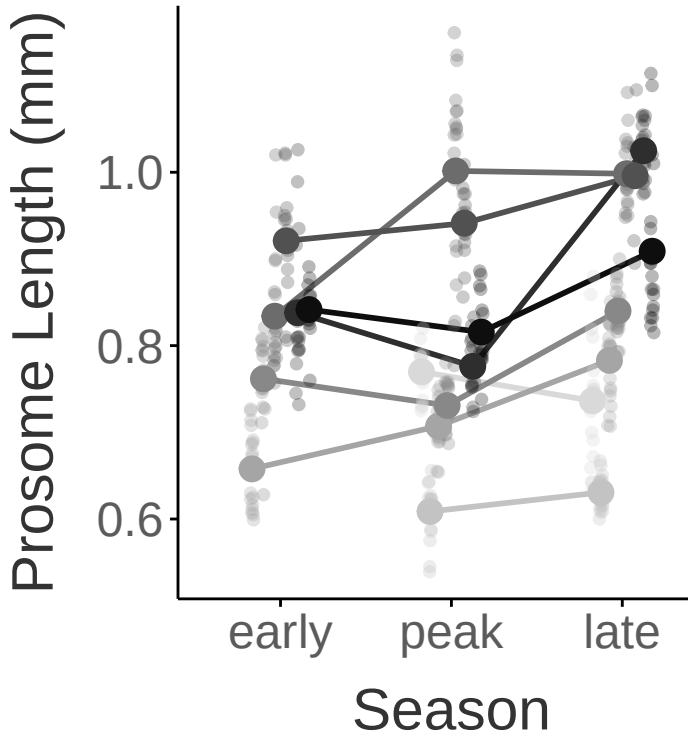
site	early	peak	late
Manatee River	0	18	20
Ft. Hamer	0	19	19
Tyler Cove	20	19	20
Ganey's Wharf	20	20	20
Esker Point	18	19	20
Sawyer Park	14	17	8
St. Thomas de Kent Wharf	15	19	19
Ritchie Wharf	16	18	19

Shown below are the measured CTmax values. Individual measurements are shown in small points for each collection. The large points indicate the mean values for each collection.

Body Size

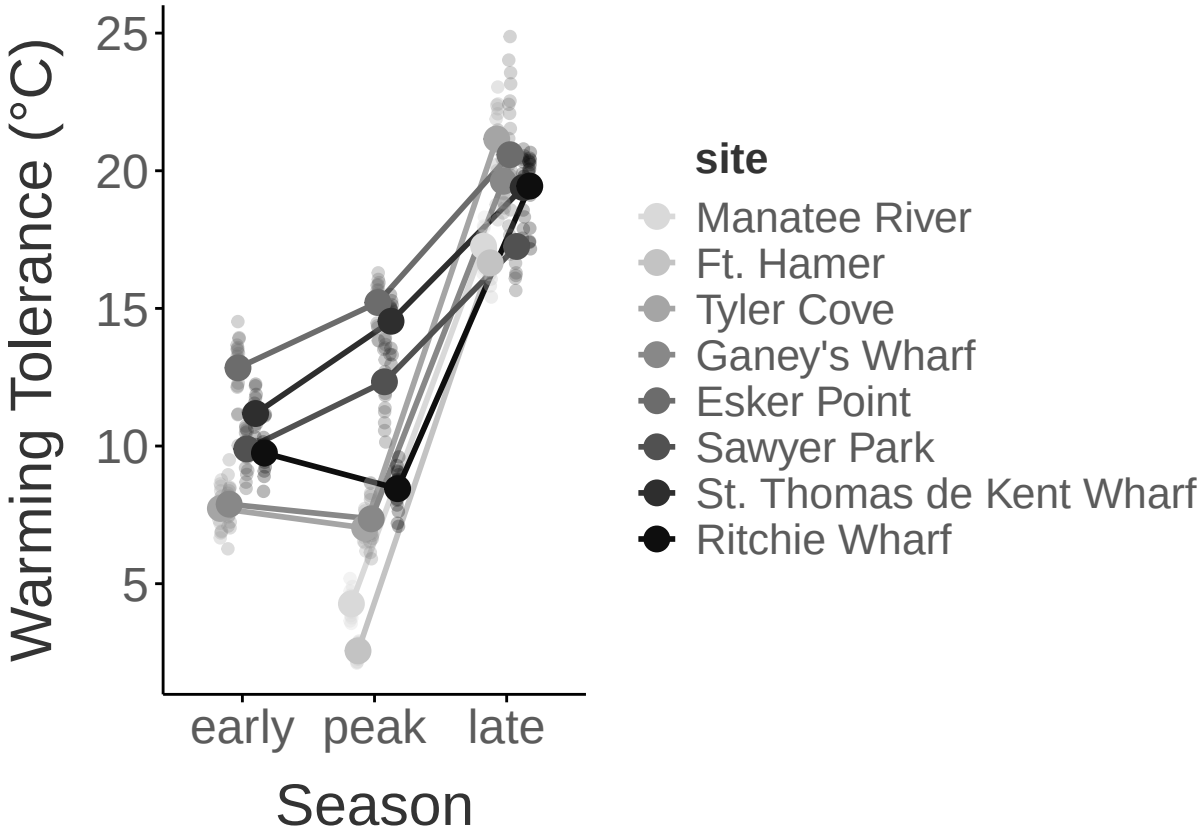
Following the CTmax assay, individuals were photographed for body size measurements. Prosome lengths were measured from these photographs using a scale micrometer and the software ImageJ. These measurements are shown below. As before, large points indicate the mean body size. While less cohesive than CTmax, a general trend of increasing body size with latitude and time of year can be seen.





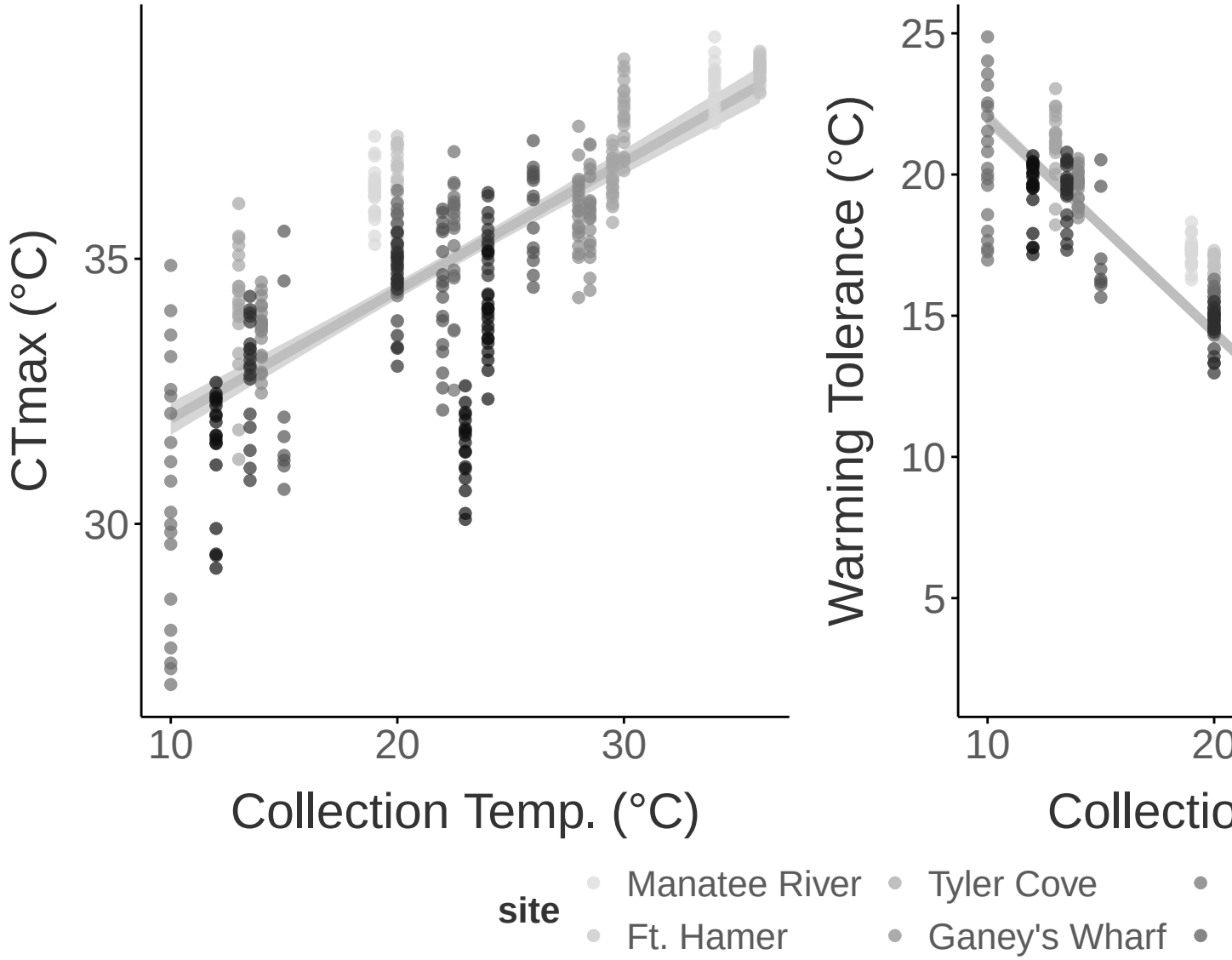
Warming tolerance

Warming tolerance (the difference between thermal limits and environmental temperatures) is a commonly used metric of climate vulnerability. We calculated this as the difference between measured CTmax values and the collection temperature. Smaller warming tolerance values indicate that populations were nearer to their upper thermal limits, suggesting greater vulnerability.

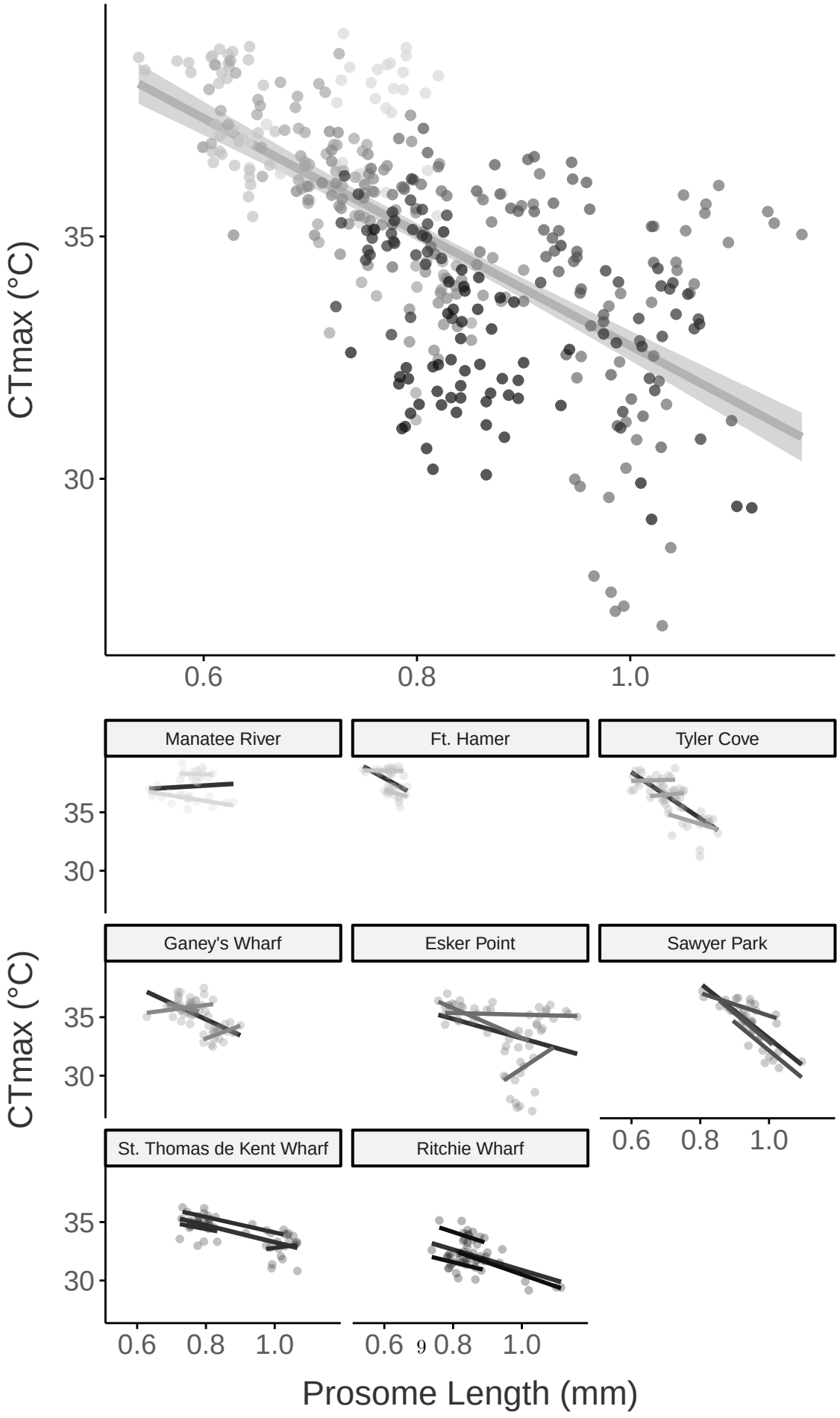


Trait Correlations

Regardless of the underlying mechanism (genetic differentiation or phenotypic plasticity), we expect that collections from warmer waters should yield copepods with higher thermal limits and smaller body sizes. Our observations largely fit this expectation, with strong increases in CTmax at higher temperatures, and a general decrease in prosome lengths as temperature increased. There was a larger correlation between CTmax and temperature than size and temperature. There are myriad reasons this might be observed, but one potential is the potential for acclimation of thermal limits. Body size is fixed in the adult stage, while CTmax can continue to vary as water temperature changes. This is indirect evidence for the importance of rapid acclimation in the observed patterns.



The relationship between prosome length and CTmax is also of interest. Given that larger body sizes are associated with cold adaptation/acclimation, it is unsurprising that there is a general trend of decreasing thermal limits with increasing size. Individual regression lines for each site are also included - the dark grey lines in the background represent the 'universal' regression for that site, with individual colored regression lines for each collection. At most sites, evidence for this negative relationship between CTmax and body size is observed across collections rather than within collections, suggesting this relationship is primaly a reflection of the response to seasonal temperature variation.

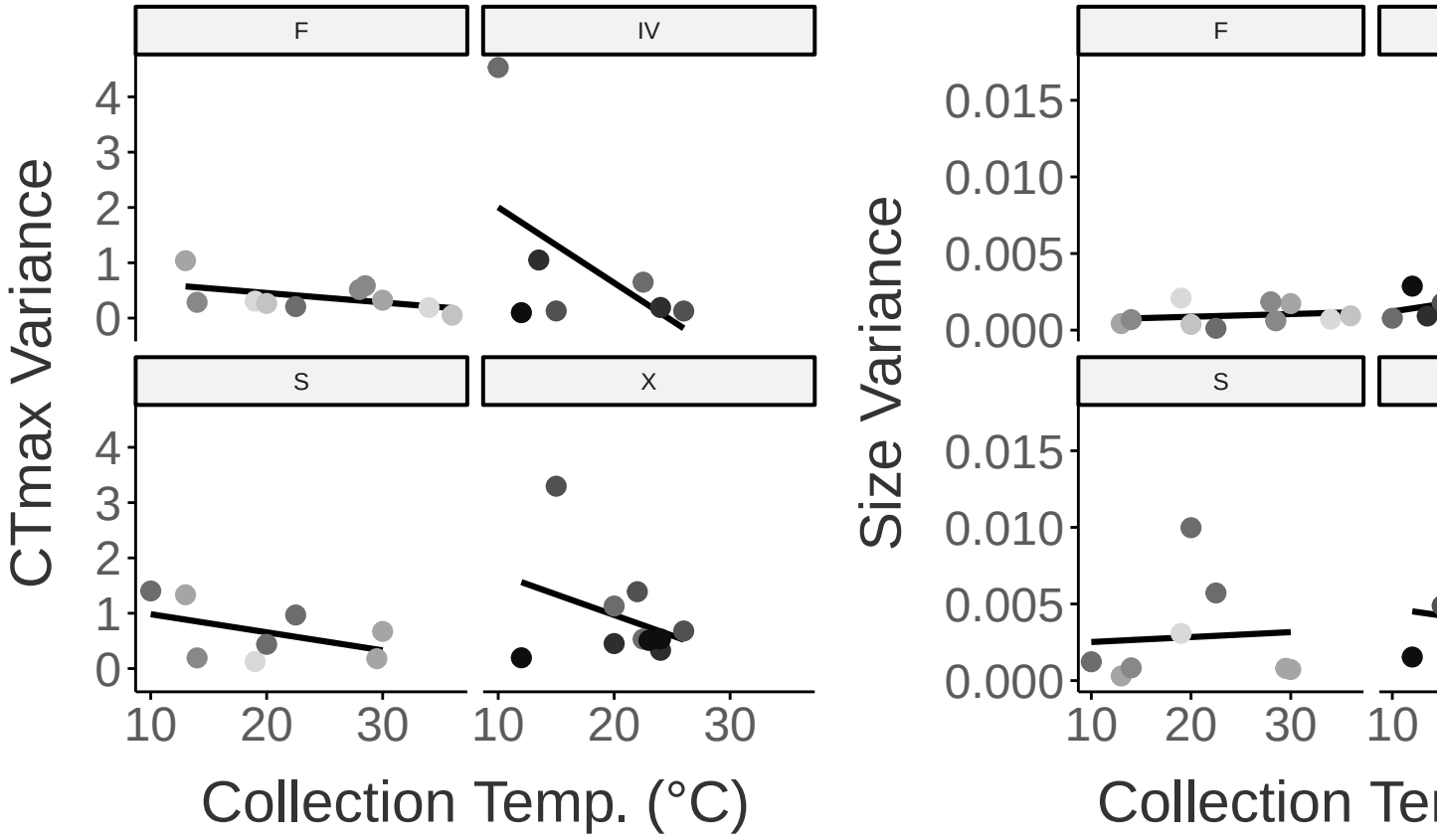

CTmax (°C)
35
30
0.6
0.8
1.0
Manatee River
Ft. Hamer
Tyler Cove
Ganey's Wharf
Esker Point
Sawyer Park
St. Thomas de Kent Wharf
Ritchie Wharf
Prosome Length (mm)

Statistical Analysis

To more formally test the relationships between CTmax, collection temperature, and size, we used a linear mixed effects model, structured as `ctmax ~ clade * temp_cent + size + (1 + temp_cent|site) + (1|tube)`. This examines the effects of clade, temperature and size on CTmax, along with clade-specific effects of temperature. Collection temperature was centered before model fitting. The model also includes random intercepts for each site and the tube number (the position of each individual within the water bath during the CTmax assay).

Trait Variability

Shown below is the trait variation (ranges) for each site. Ranges are calculated for each collection separately.



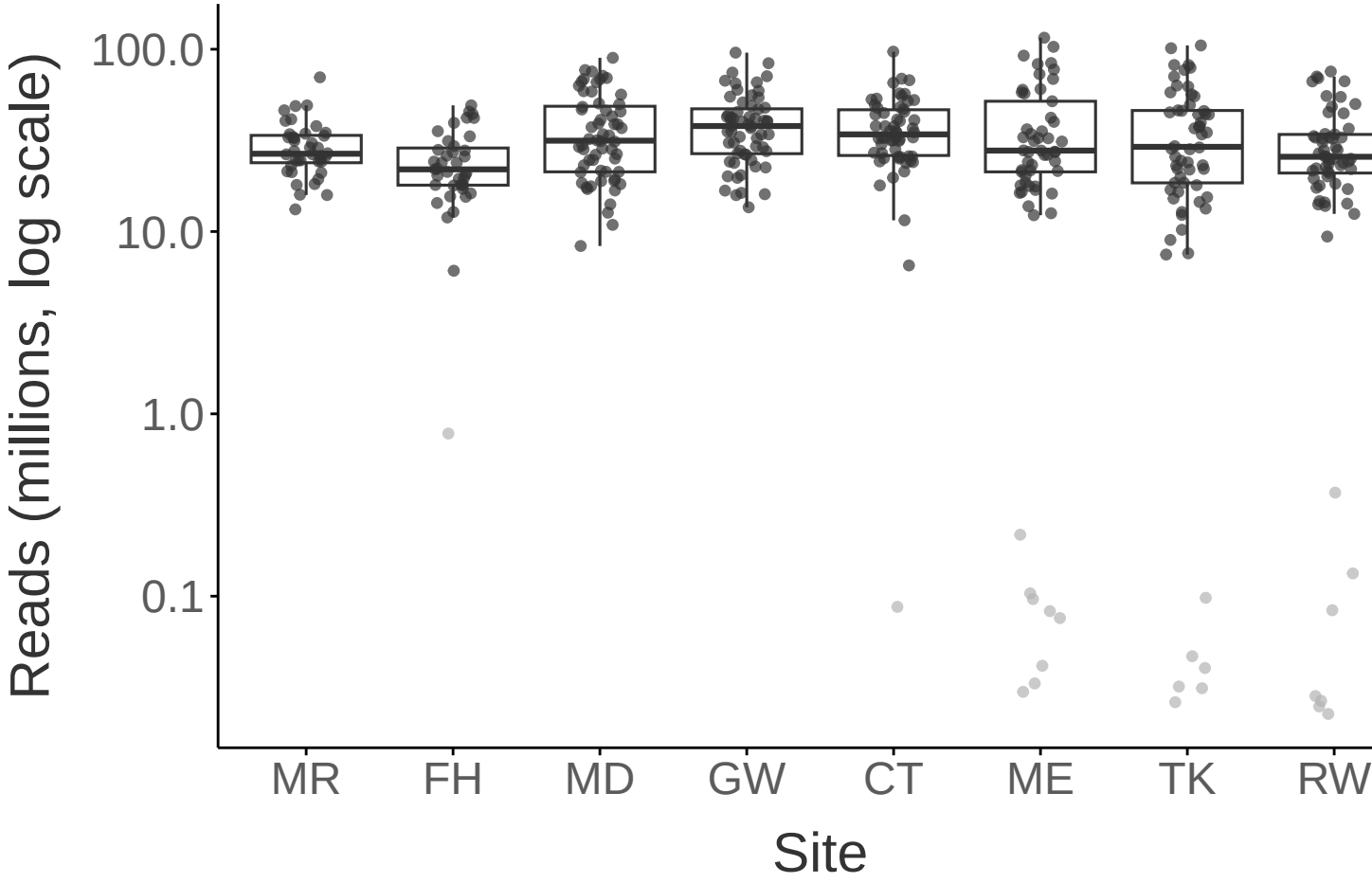
Changes in trait variance may be indicative of phenotypic selection. If selection (as opposed to acclimation) are driving seasonal changes, we may expect to see a reduction in variance in the peak samples relative to the early season samples. Note that early season collection temperatures this year were higher than expected, driven by fairly strong heatwaves across the North Atlantic. As shown in the temperature profiles for each site, the 'early' samples were collected just after high temperatures were reached, while 'peak' samples were collected after sites had experienced high temperatures for several weeks (generations). As warming tolerances were fairly high throughout this period, we will assume that selection was weak before the early samples. If the early onset of high temperatures filtered out vulnerable genotypes prior to our sampling, the results will be a conservative estimate of the effects of selection on trait variance.

Low-Coverage Whole Genome Sequencing

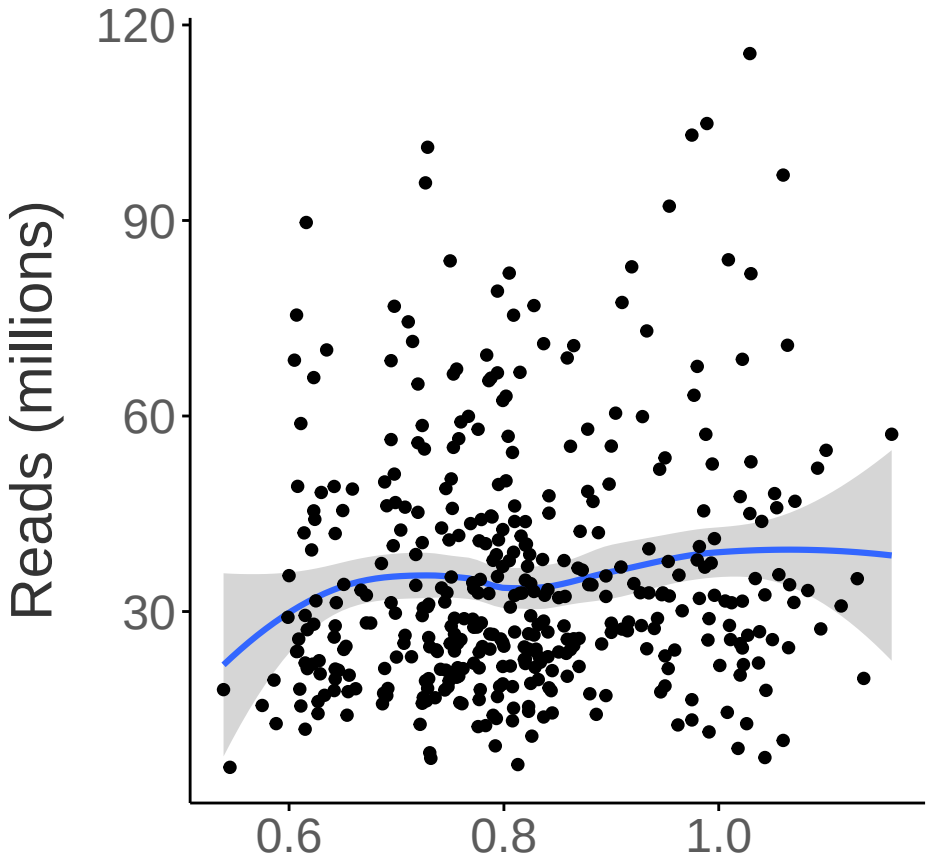
While plasticity appears to play a large role in the response of *Acartia tonsa* populations to seasonality, a growing number of studies have also uncovered patterns in allele frequency indicating of rapid adaptive

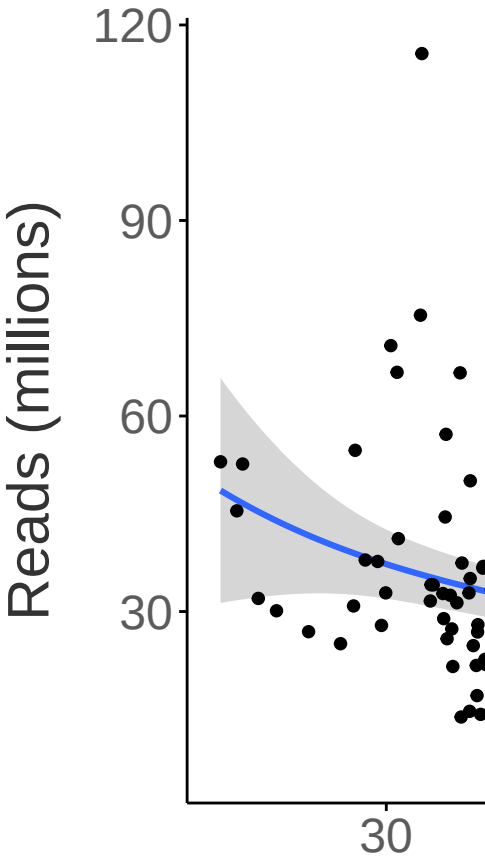
responses to seasonality. Our data set of seasonal samples across a large latitudinal gradient enables a comparison of these fine scale temporal patterns with the larger latitudinal patterns in allele frequency to determine whether the same alleles driving rapid seasonal adaptation are in play over larger spatial (and longer temporal) scales.

After phenotyping, each individual was preserved in 95% ethanol. Individual DNA libraries were prepared using Twist Bio 96-plex prep kits, then sequenced on two lanes of an Illumina NovaSeq X Plus 25B by Novogene. Reads were de-multiplexed first to the plate level, and then to the sample level. Read processing, mapping and QC were done on the Unity cluster (Snakemake project `aton_lcwg`); the numbers below come from the exported QC tables (`results/00_qc/sample_summary.tsv`, `coverage_per_sample.tsv`) joined in `Scripts/03_sample_table.R`. Failed libraries (`library_failed`) are shown in grey.

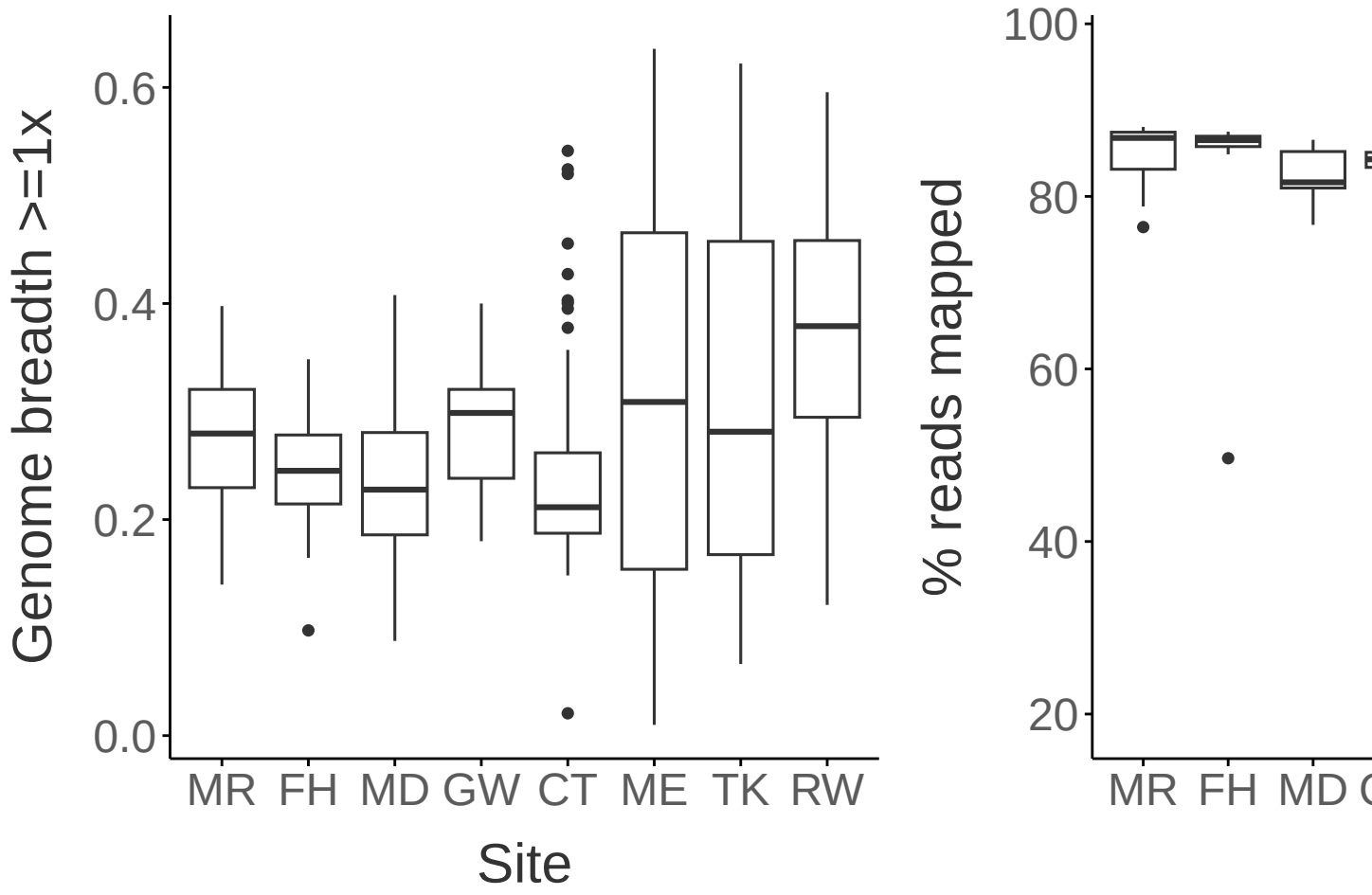


The number of reads per individual varies, but there is no clear relationship between sequencing depth and individual size or CTmax (phenotype analysis set only).



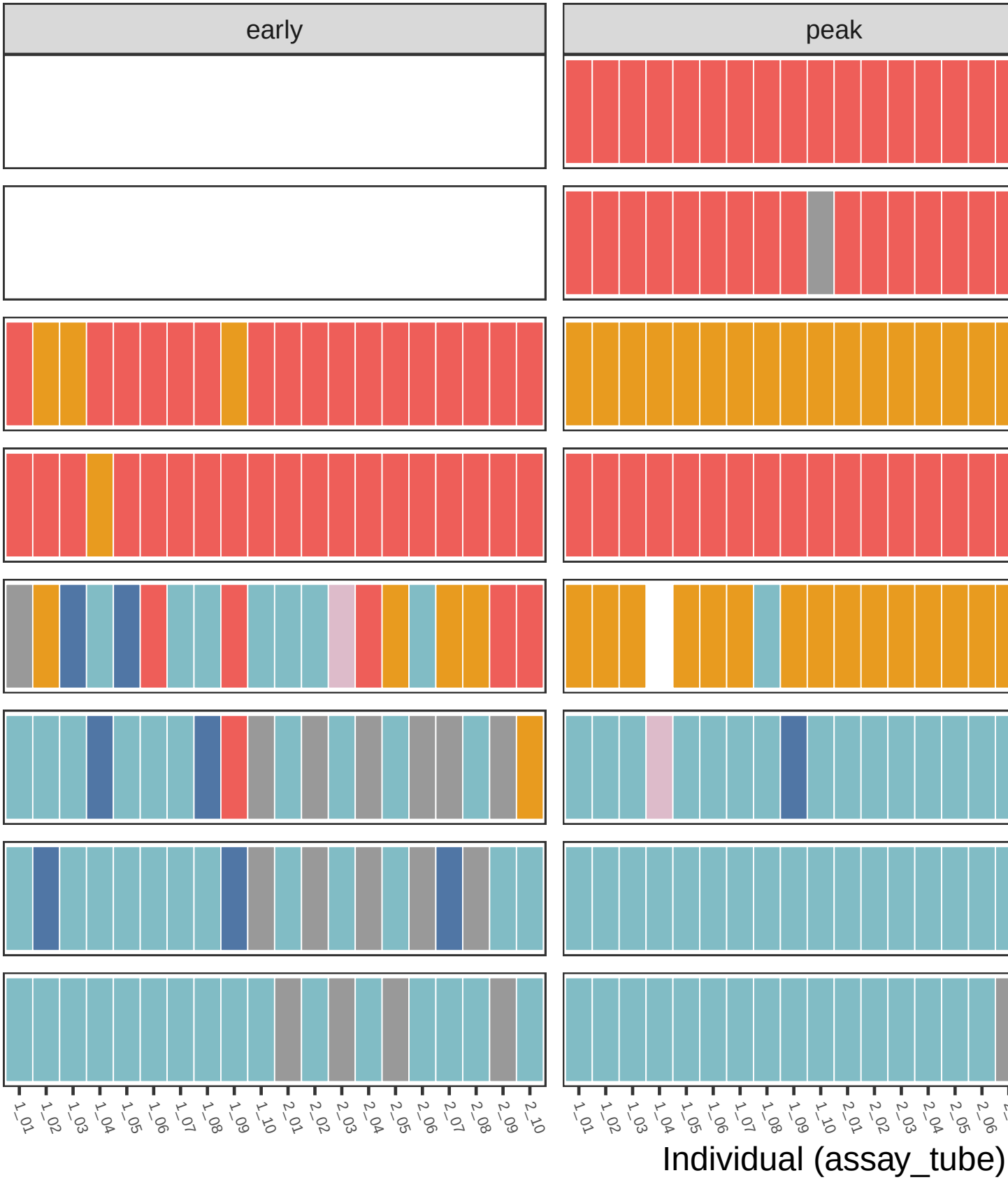


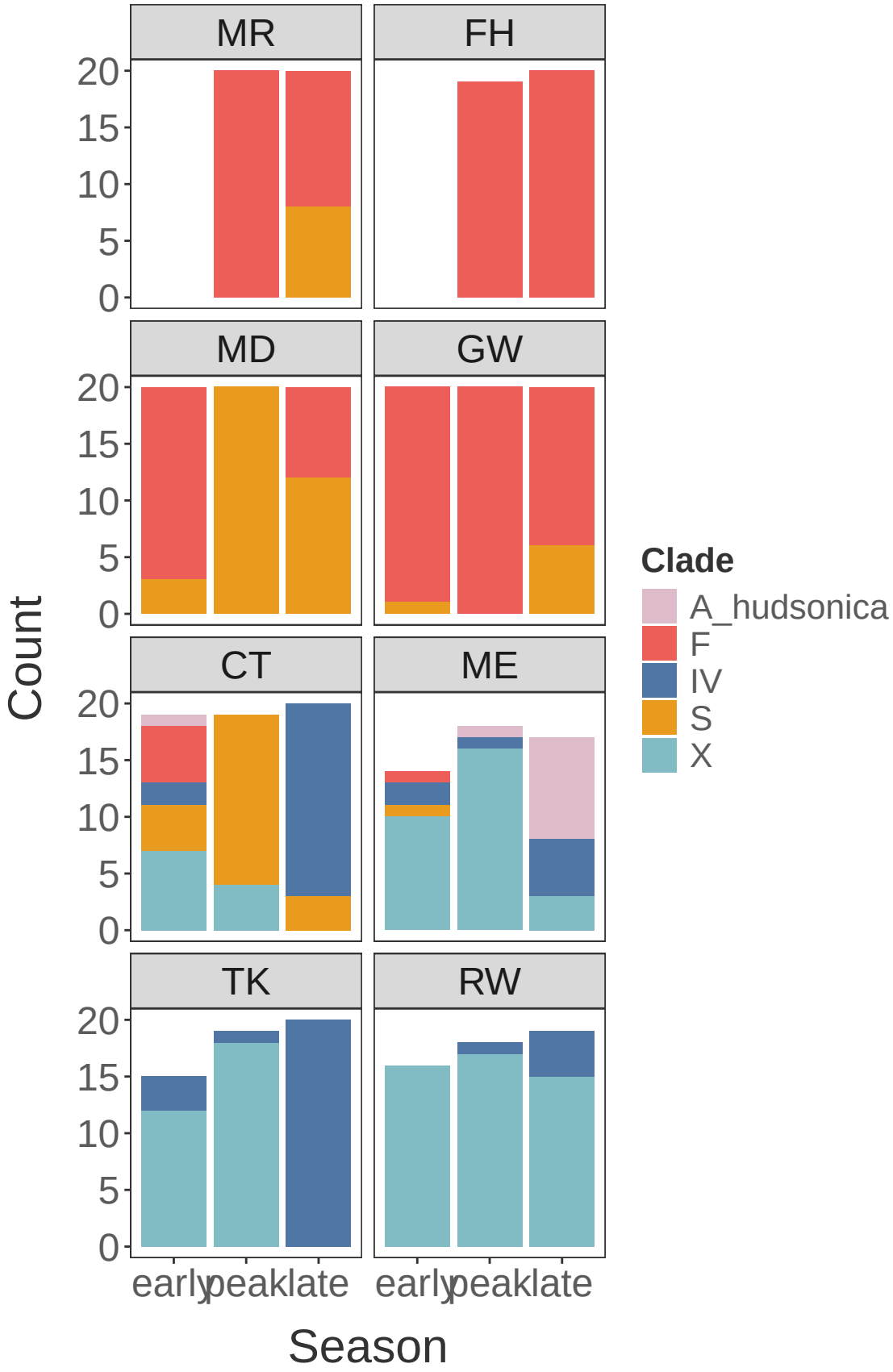
	Sum Sq	Df	F value	Pr(>F)
size	9.988975e+14	1	2.7797642	0.0962568
ctmax	3.807035e+11	1	0.0010594	0.9740508
Residuals	1.415824e+17	394	NA	NA



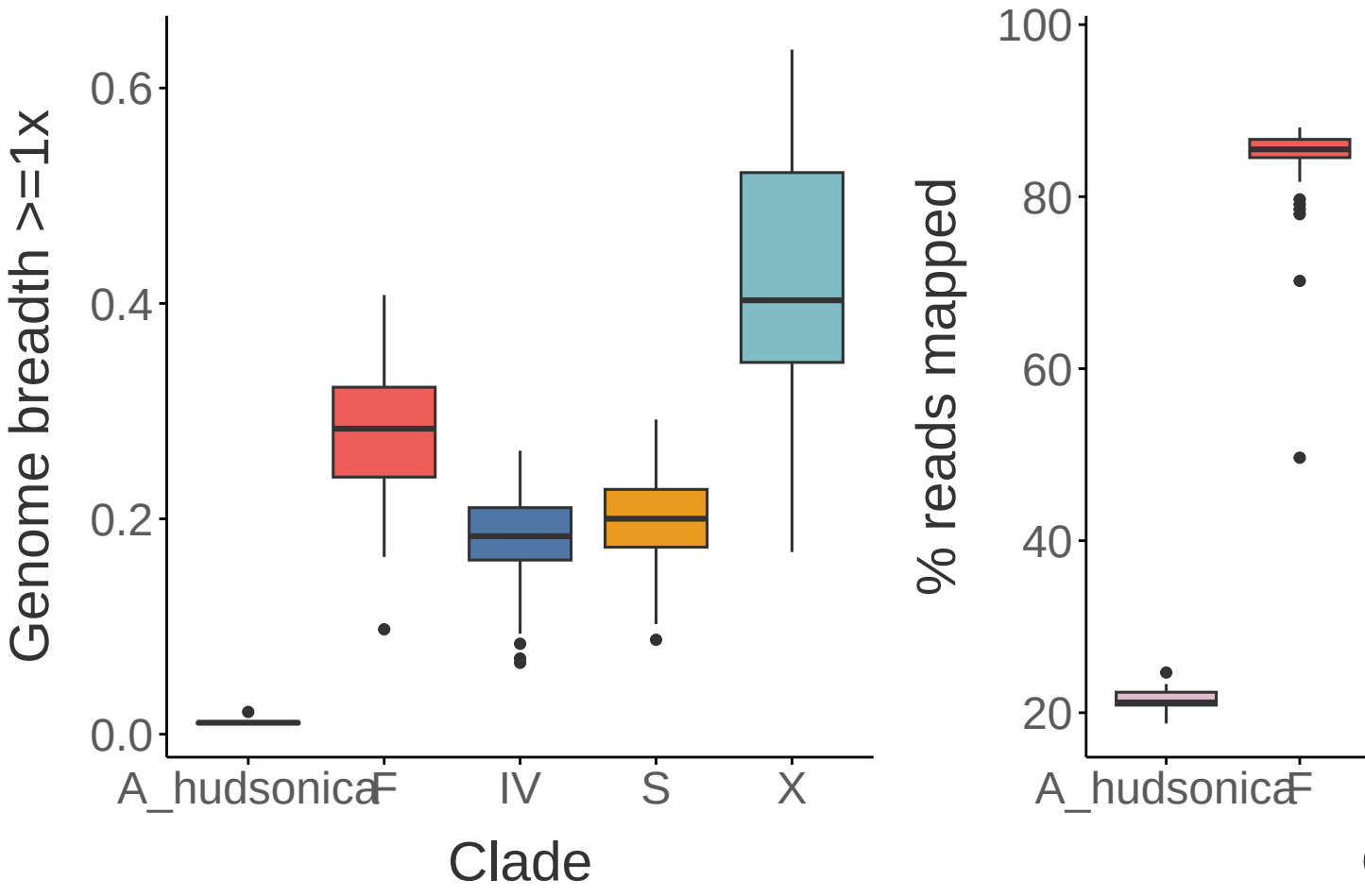
Clade IDs

Acartia tonsa is well known to comprise multiple, deeply diverged clades. Clade calls currently come from prior_coi (`prior_coi` = COI reads skimmed from the low-coverage data; `stage1` = final calls from the cluster, mitochondrial COI tree plus nuclear clustering). Calls from failed libraries are not used.


early
peak
1_01
1_02
1_03
1_04
1_05
1_06
1_07
1_08
1_09
1_10
2_01
2_02
2_03
2_04
2_05
2_06
2_07
2_08
2_09
2_10
Individual (assay_tube)

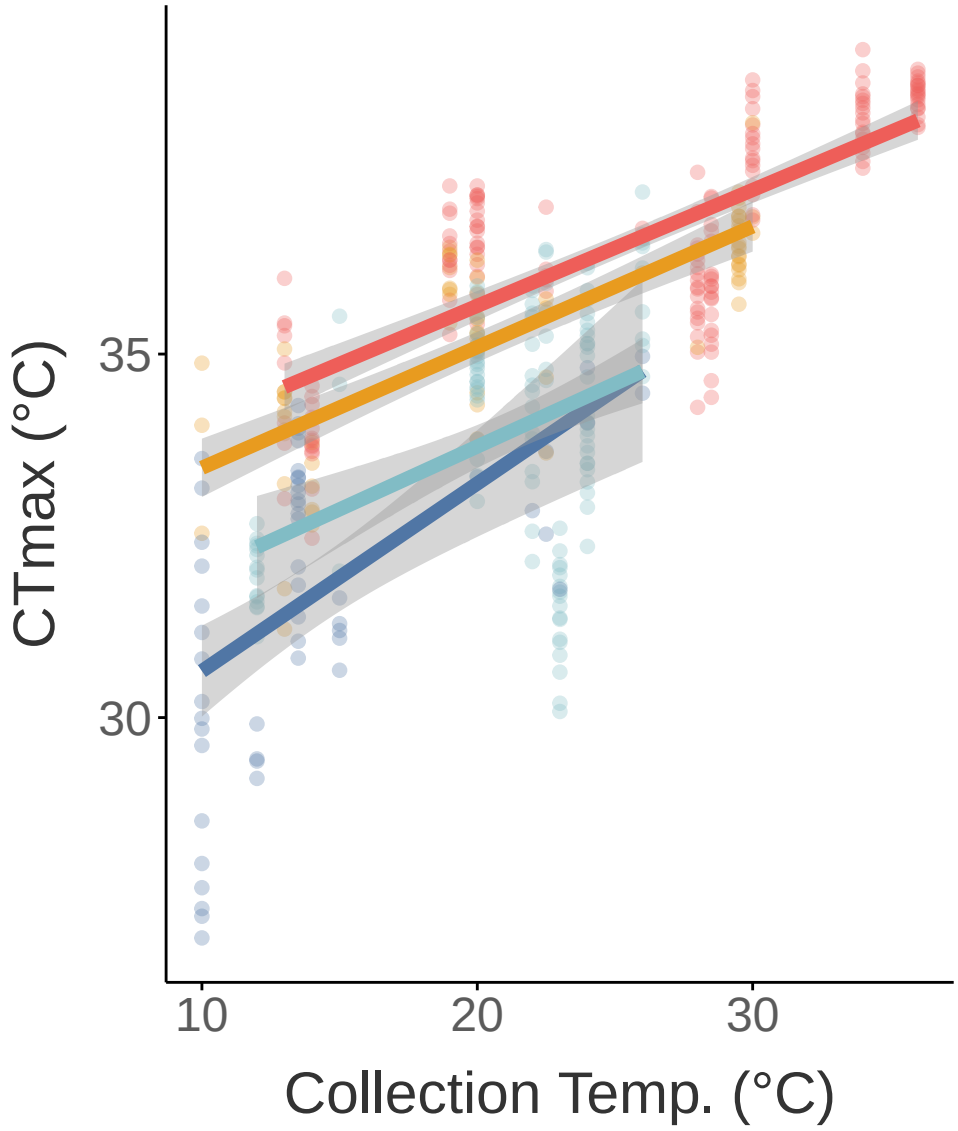

MR
FH
MD
GW
CT
ME
TK
RW
Count
0
5
10
15
20
early
peak
late
Season
Clade
A_hudsonica
F
IV
S
X

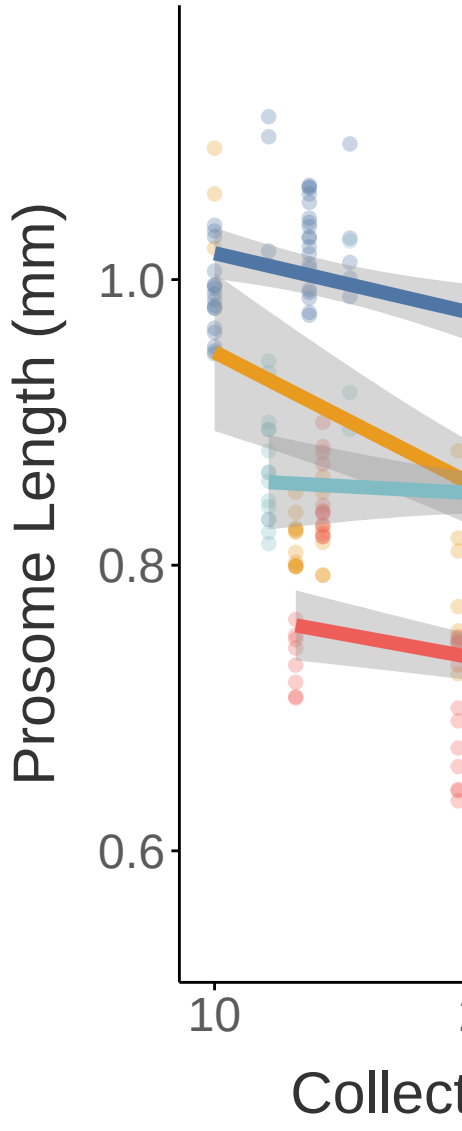
These clades are deeply diverged, which affects the alignment and final coverage of the genome data (clade X is closest to the reference genome).

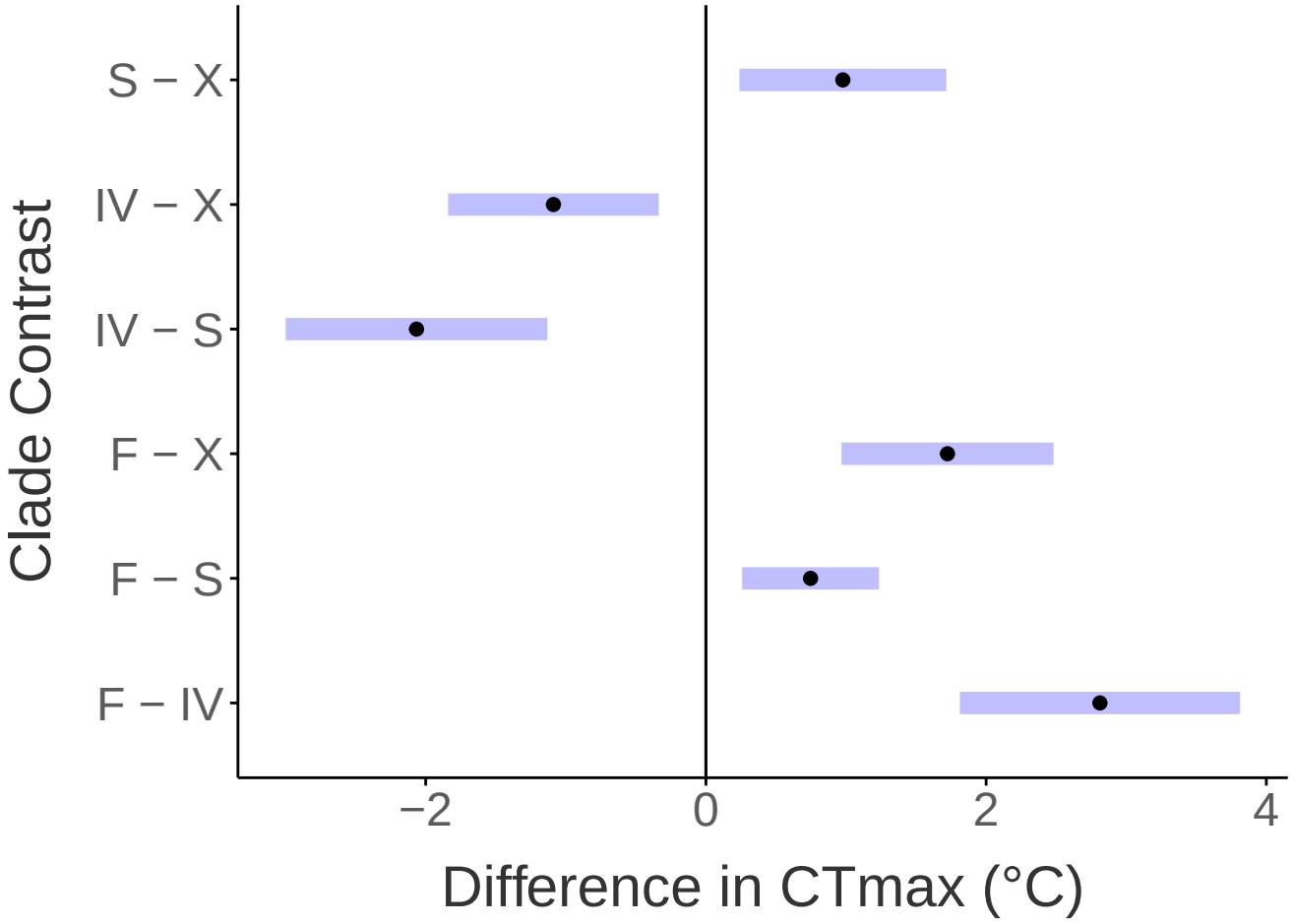


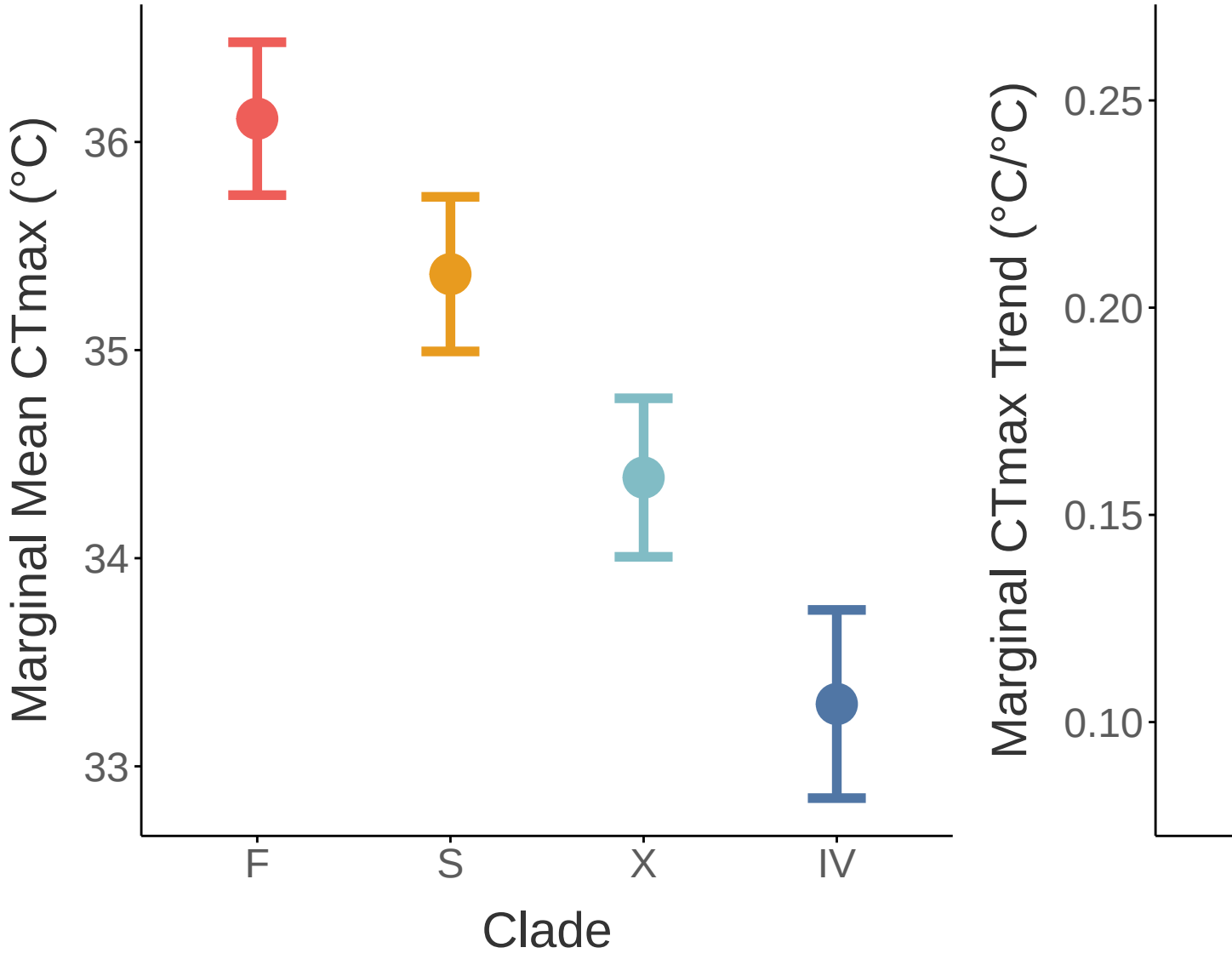
Clade-specific Physiology

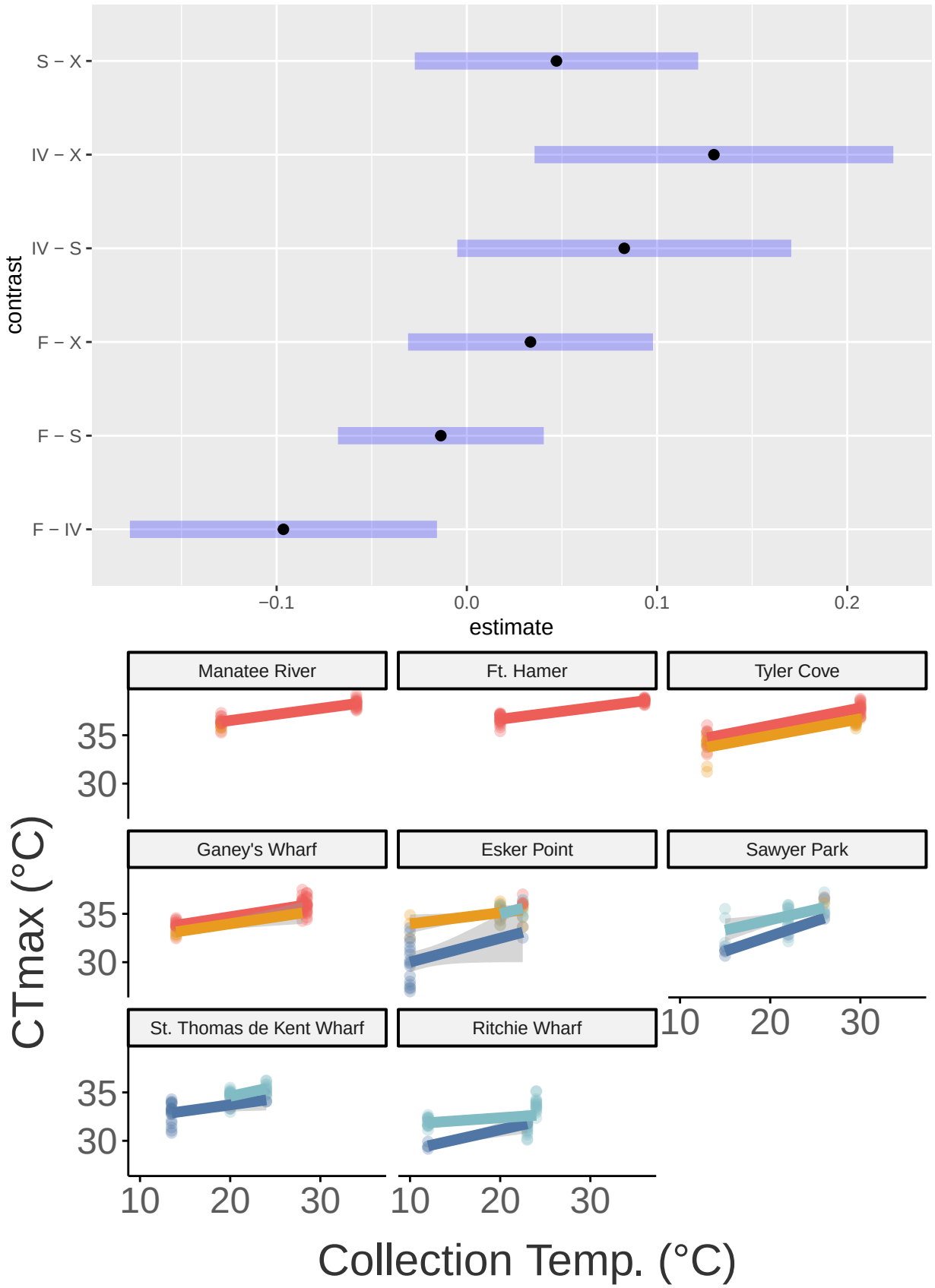
These COI clades were then merged with the CTmax and body size data set to examine clade-specific patterns of divergence. By plotting CTmax and size against the collection temperature we can see that these clades appear to be physiologically differentiated. Clades F and S tend to have higher CTmax, even when collected at the same temperatures as Clades X and IV. Size is more variable, with varying degrees of temperature sensitivity across the clades. Overall, however, Clade F tends to be smaller than Clades S and X. Clade IV tended to be the largest.


CTmax (°C)
35
30
10
20
30
Collection Temp. (°C)


Prosome Length (mm)
1.0
0.8
0.6
10
Collect

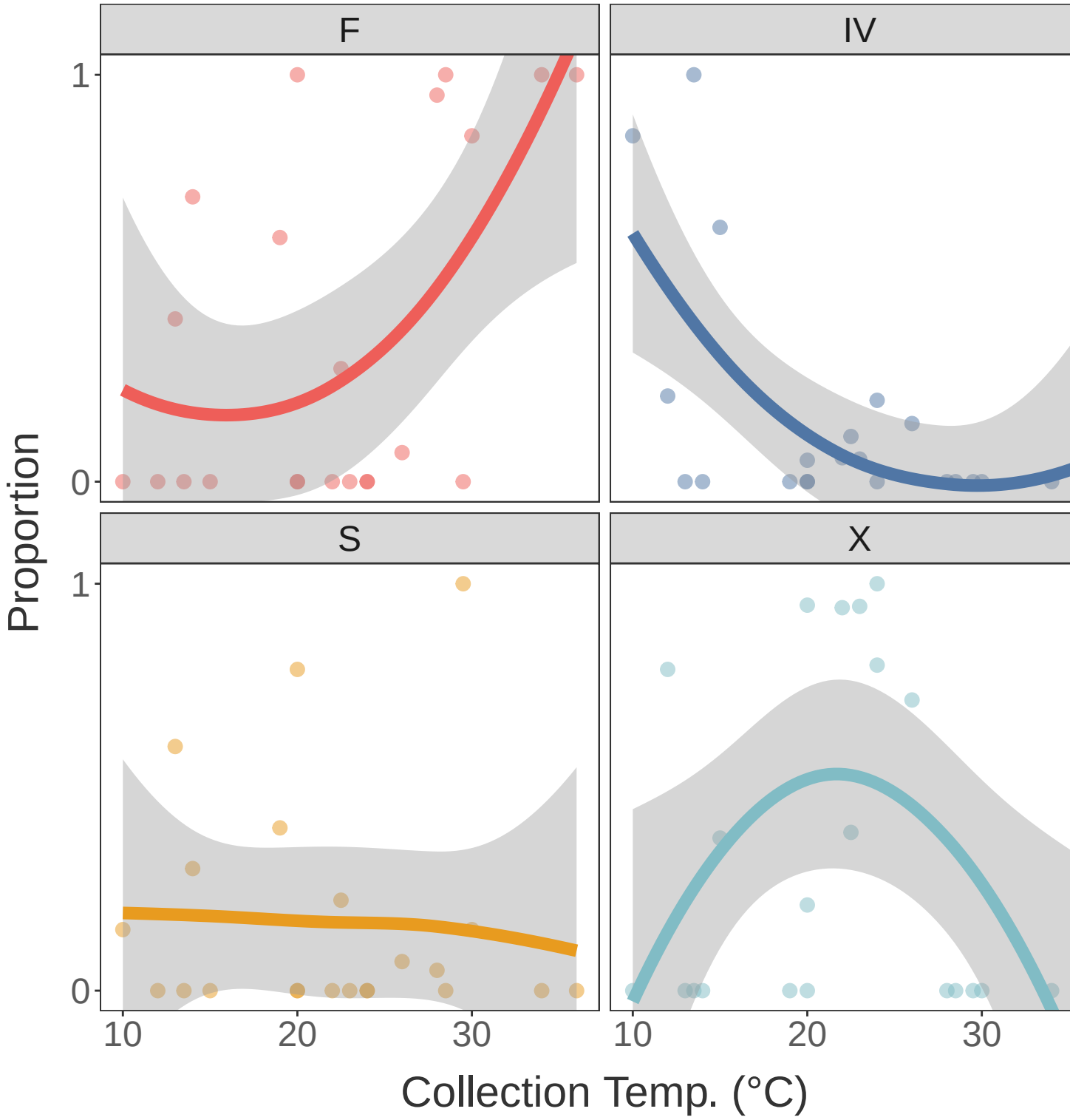

Clade Contrast
S − X
IV − X
IV − S
F − X
F − S
F − IV
−2
0
2
4
Difference in CTmax (°C)

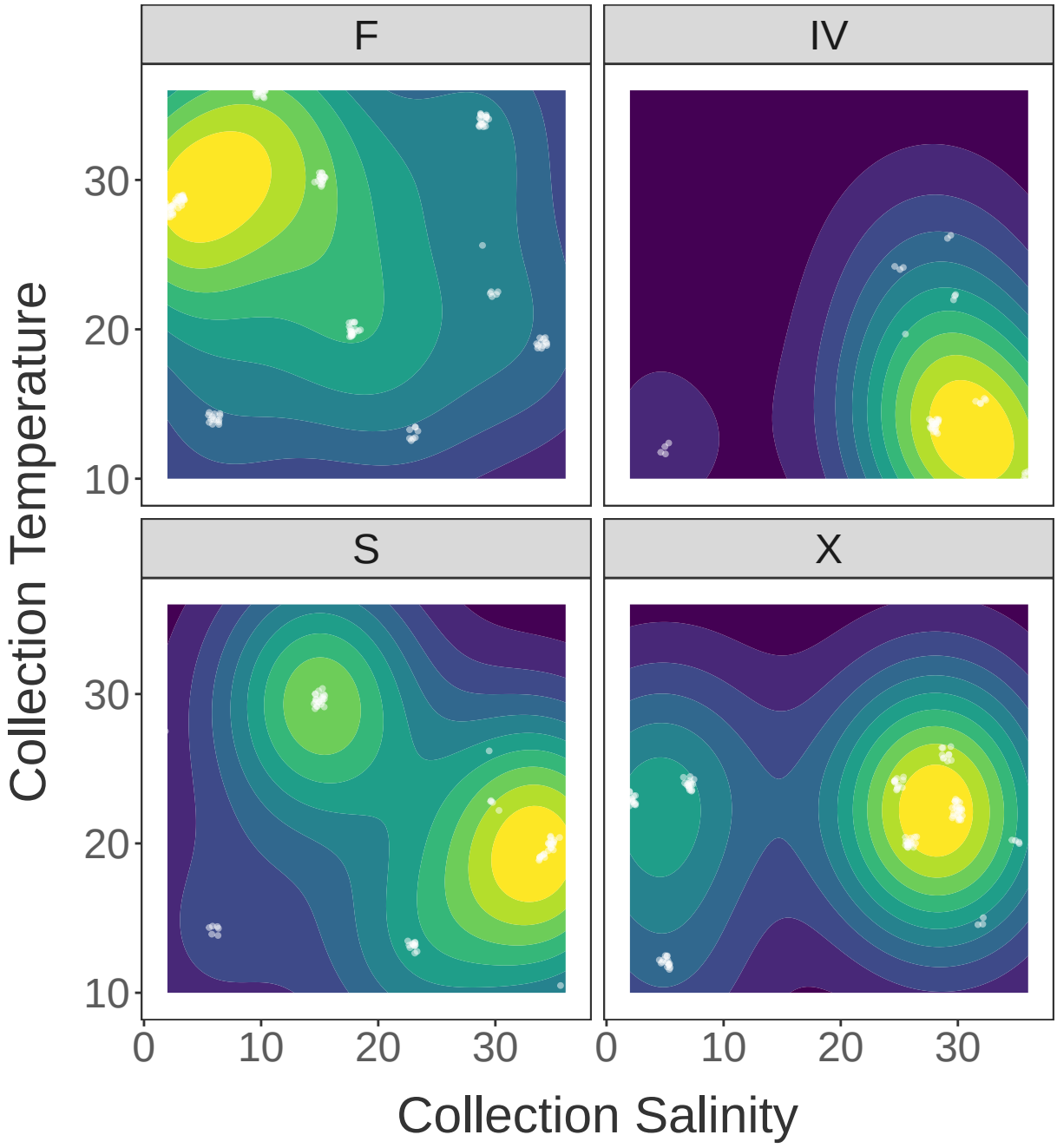

Marginal Mean CTmax (°C)
36
35
34
33
F
S
X
IV
Clade
Marginal CTmax Trend (°C/°C)
0.25
0.20
0.15
0.10


S − X
IV − X
IV − S
F − X
F − S
F − IV
contrast
−0.1
0.0
0.1
0.2
estimate
Manatee River
Ft. Hamer
Tyler Cove
Ganey's Wharf
Esker Point
Sawyer Park
St. Thomas de Kent Wharf
Ritchie Wharf
35
30
10
20
30
CTmax (°C)
Collection Temp. (°C)

The general result here is that Clade F appears to be more thermally tolerant than Clades S and X, which in turn are more thermally tolerant than Clade IV. Clade IV does however, appear to exibit stronger plasticity than the other clades (although this pattern is only marginally significant in two of the comparisons; IV-S and IV-X)

These differences in physiology correspond to the distributional patterns for each clade. Using a Loess smoother (span = 2) highlights how the proportional abundance of the clades is affected by temperature in different ways. Clade F occurs in high proportion at high temperature, Clade IV occurs in high proportions at low temperatures, while Clade X occurs in high proportions at intermediate temperatures. Clade S generally occurs in low proportions, with a few exceptions.


F
IV
S
X
Proportion
1
0
10
20
30
Collection Temp. (°C)



PCA

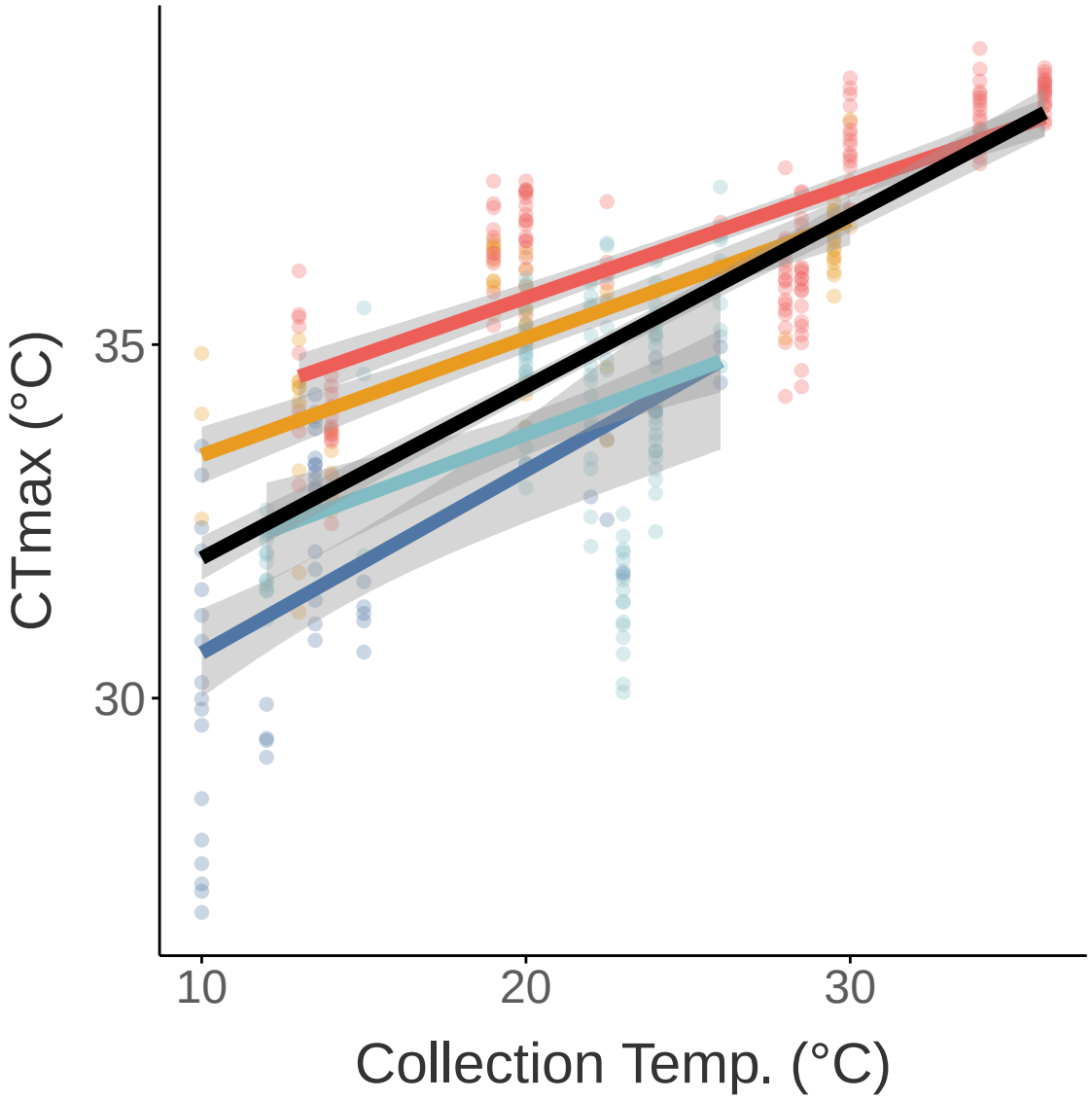
Per-clade PCA (PCAngsd) and admixture (NGSadmix) come from Stage 1 of the cluster workflow (`results/04_structure/`). They will be read here from the cluster export once Stage 1 is done; the earlier exploratory PCA code (based on covariance matrices from before the Snakemake pipeline) is kept in `Output/Reports/legacy/report_pre_migration.Rmd`.

Partitioning Vulnerability

Our study examines spatial and temporal variation in upper thermal limits for field-acclimated individuals. Given the short generation time of these copepods, our results likely reflect the combined influence of phenotypic plasticity (acclimation to the changes in temperature), local adaptation (variation amongst the

sites), and genetic variation between the clades.

The four clades, while morphologically difficult to differentiate, have distinct geographic distributions and physiological tolerances. The inclusion of this clade-level variation decreases the slope of the CTmax vs. temperature regression. Ignoring this source of variation would lead to an overestimated capacity for *A. tonsa* to respond to warming (either through acclimation or via gene flow and evolutionary rescue).



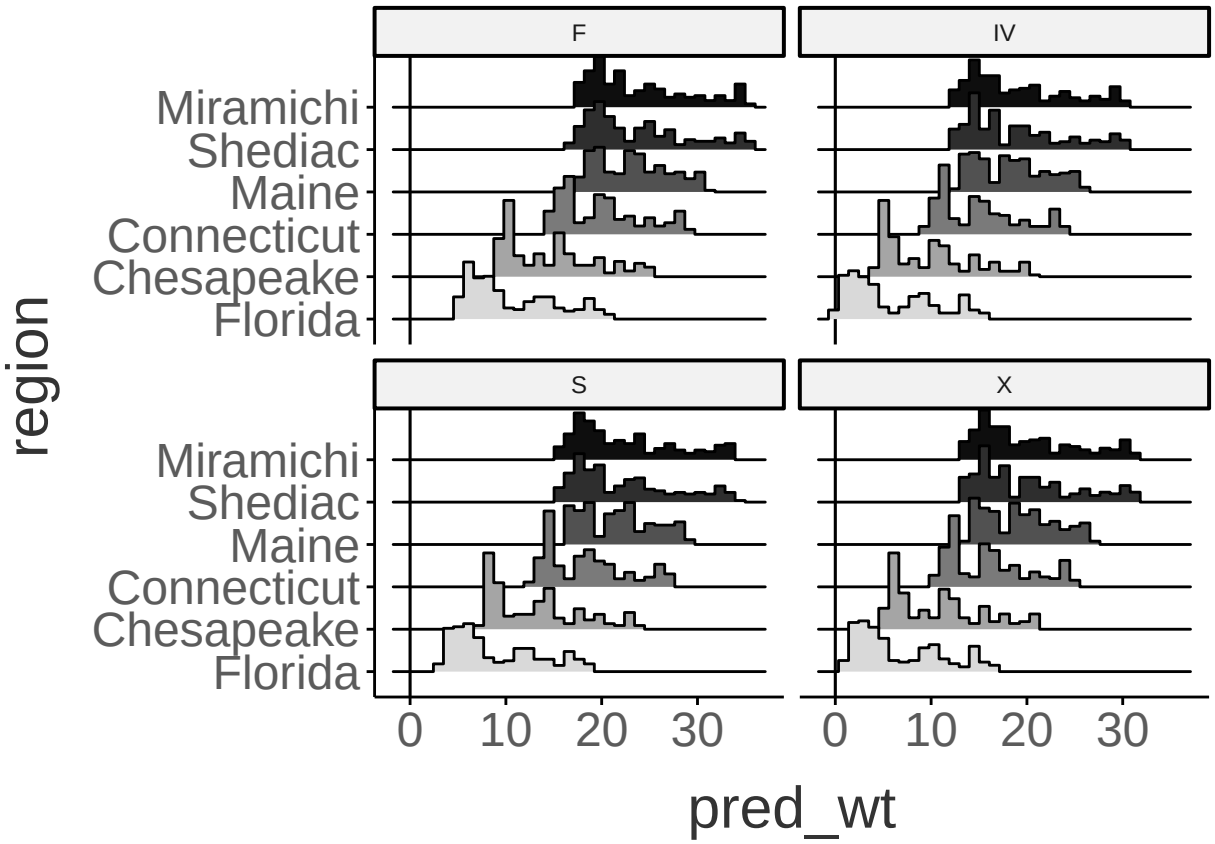
We illustrate the effects of these different levels of variation by calculating warming tolerance for each collection (site x collection temperature combination) under three different scenarios.

Scenario 1 (Fixed CTmax)

Warming tolerance was calculated using the average CTmax from the peak season collections (across all sites and individuals regardless of clade). This essentially assumes a fixed CTmax that is constant across the geographic range examined.

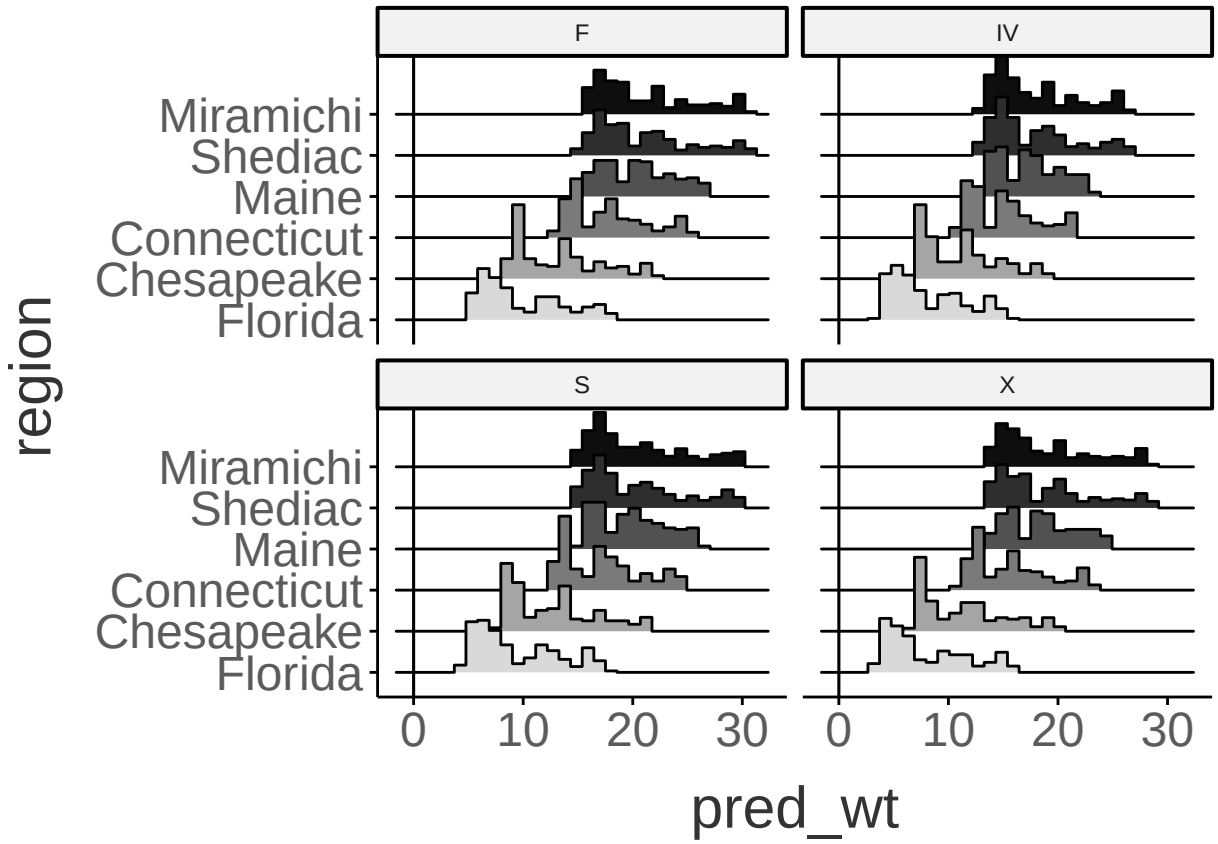
Scenario 2 (Clade-level variation)

Here, the same approach is taken but for each clade separately. The average CTmax from the peak season for each clade was used as a fixed CTmax value when calculating warming tolerance.



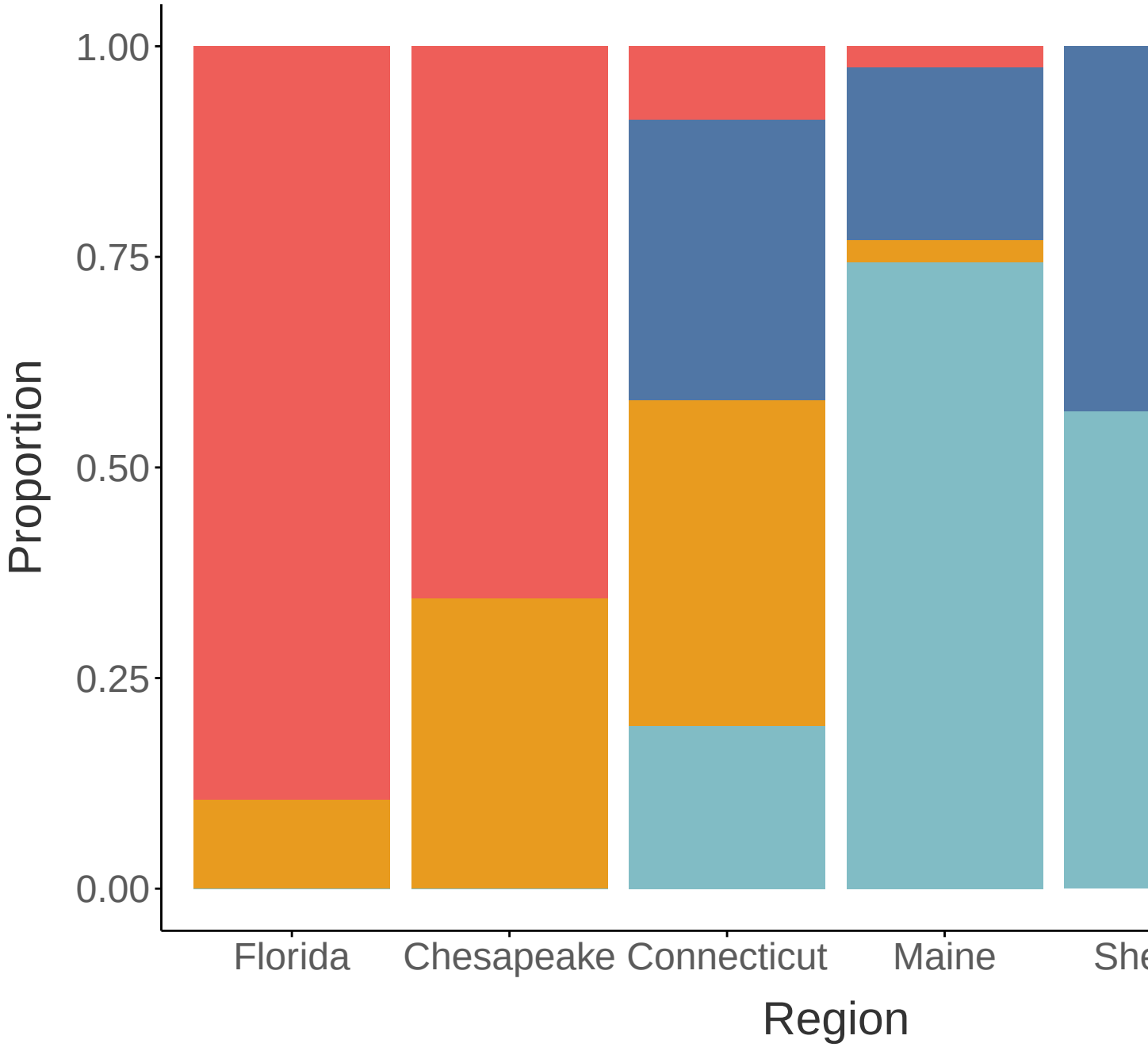
Scenario 3 (Clade variation and acclimation/adaptation)

Here CTmax is predicted based on the continuous temperature data and a model of CTmax as a function of the interaction between collection temperature and clade. The predicted CTmax values fall within the range of CTmax values observed in the field.



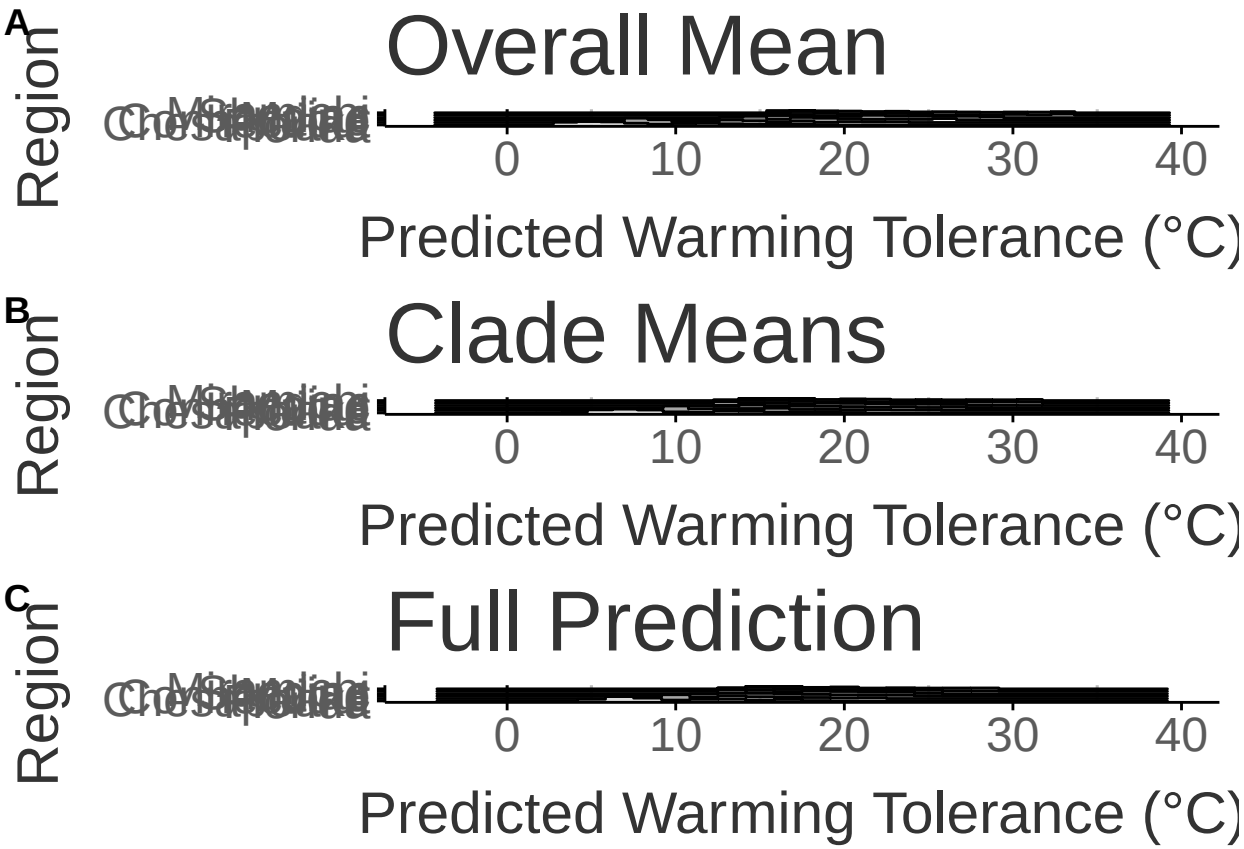
Comparing Scenarios

There is strong geographic variation in the distributions of these clades. It does not, for example, make sense to examine warming tolerance of Clade X individuals in Florida, as this clade was not collected from lower latitudes. To incorporate this distribution information, simulated populations were re-assembled reflecting the proportional clade composition within each region (calculated across the entire set of collections).



Warming tolerance profiles for each clade were added in these proportions to generation populations of 100 individuals.

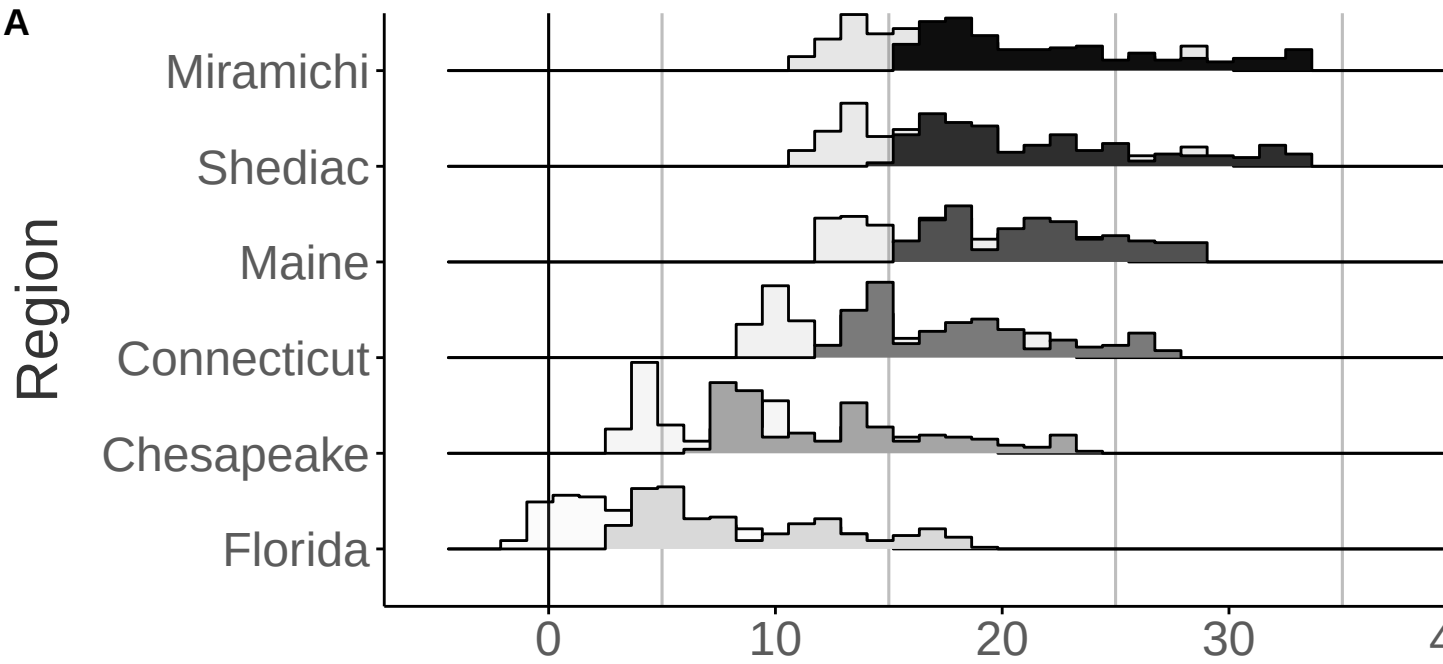
The warming tolerance distributions for each region and the three different scenarios are shown below.

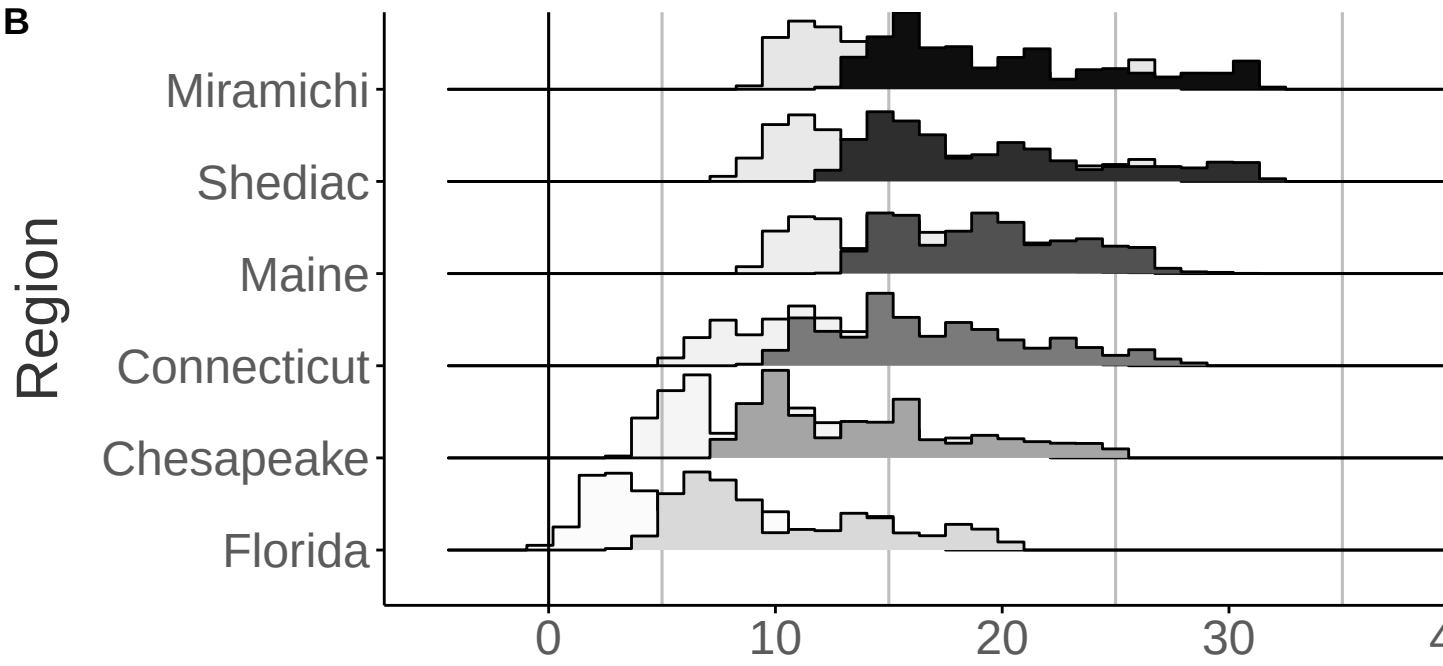


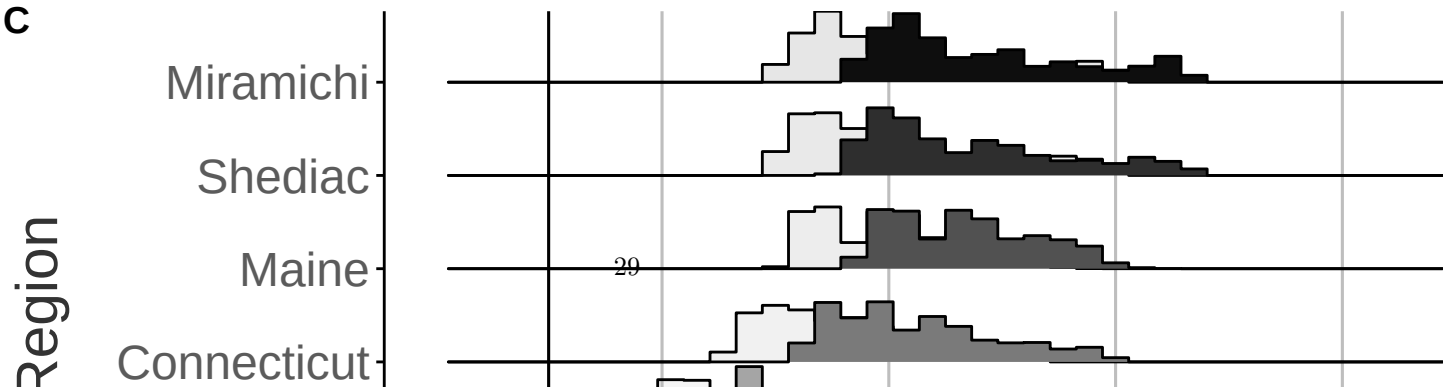
Overall, differences were fairly minor, with slight increases in predicted warming tolerance in low latitude regions and slight decreases in higher latitude regions.

We also examined potential effects of warming (a +4°C increase in temperature). These predictions assume clade composition does not change over time and that warming is constant throughout the year.

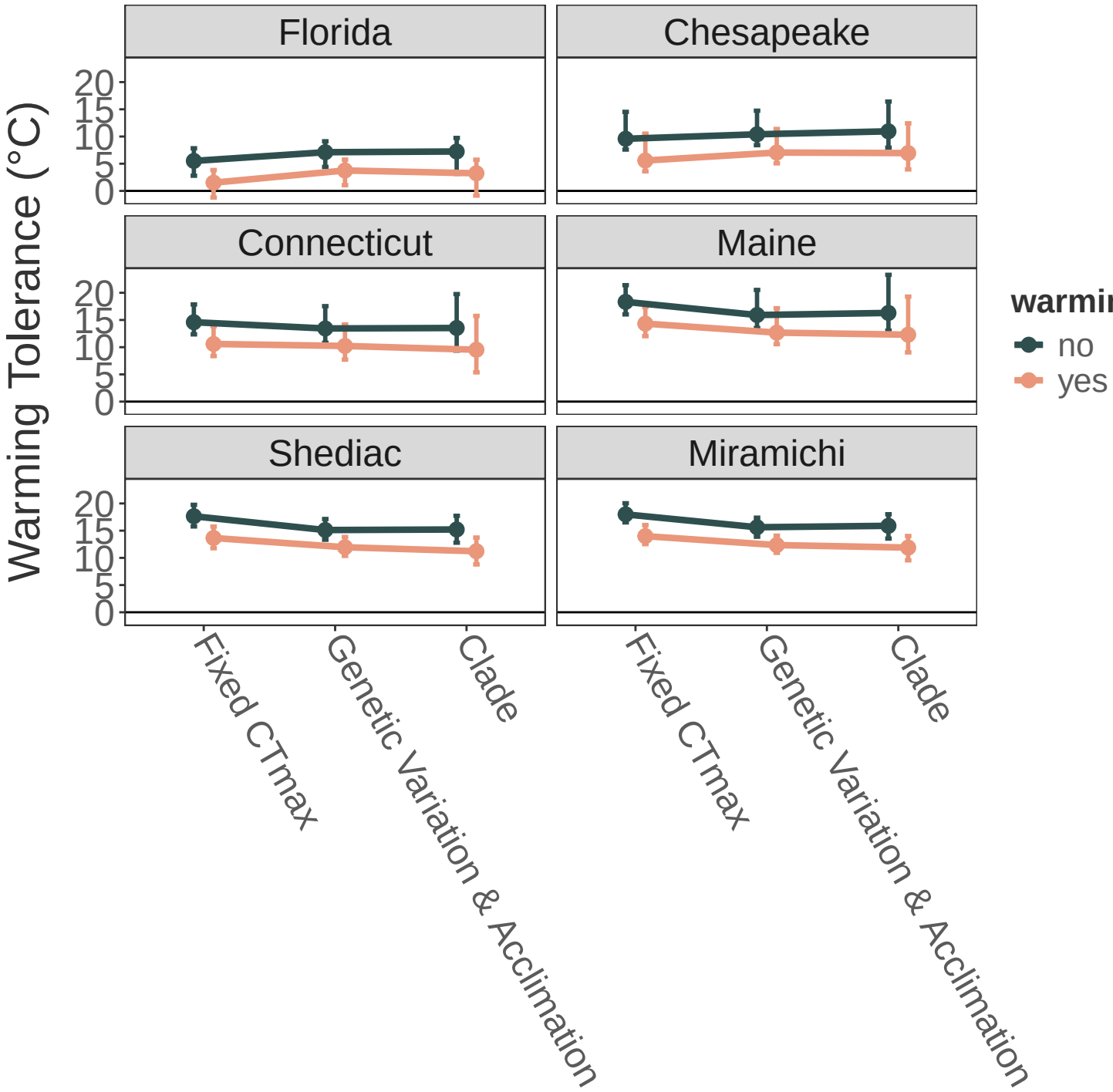
Unsurprisingly warming decreases the predicted warming tolerance. Similar patterns are observed as in the comparison between the initial three scenarios - predicted warming tolerance under warming is lowest in Florida and highest in the high latitude regions. Similarly, including clade-level variation and acclimation increases warming tolerance in the lower latitude regions and decreases it in the higher latitude regions.


A
Region
Miramichi
Shediac
Maine
Connecticut
Chesapeake
Florida
0
10
20
30


B
Region
Miramichi
Shediac
Maine
Connecticut
Chesapeake
Florida
0
10
20
30


C
Region
Miramichi
Shediac
Maine
Connecticut

To simplify the comparisons, here we subset the data to just August and September (the warmest months and the season typically associated with the highest abundance of A. tonsa).



Changes were slight between the scenarios, with the same relative patterns between the current and warming scenarios. Compared to the fixed CTmax estimate, incorporating both genetic diversity and acclimation reduced the range of warming tolerance estimates across the samples by ~30%.

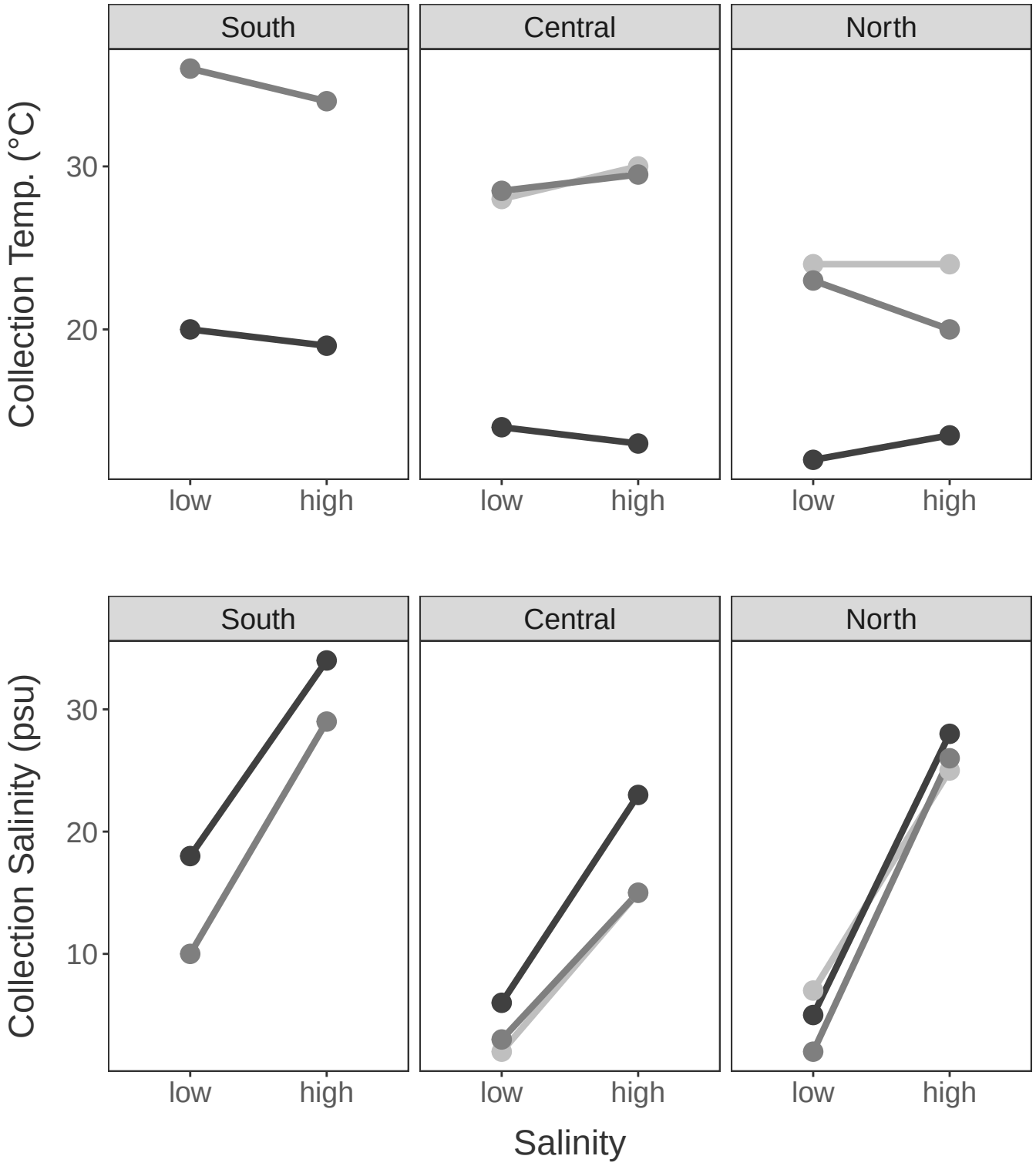
Comparison	WT Range	% Decrease
Fixed CTmax	13.05	0.00
Clade	9.72	25.53
Genetic Variation & Acclimation	8.99	31.13

Salinity Comparisons

Nested within each of the three regions (South, Central, and Northern regions) are pairs of low and high salinity sites:

Region	Low.Salinity	High.Salinity
South	Ft. Hamer	Manatee River
Central	Ganey's Wharf	Tyler Cove
North	Ritchie Wharf	St. Thomas de Kent Wharf

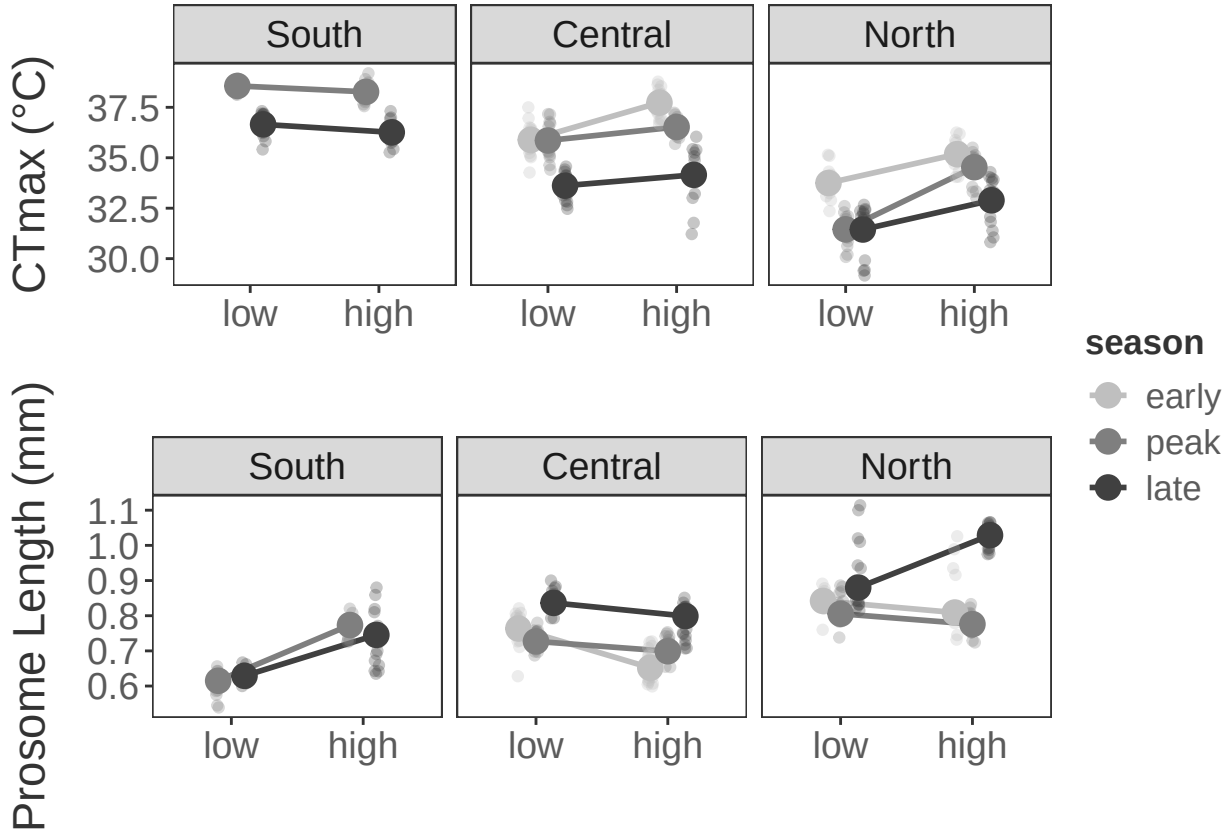
There are fairly well-established divergences between high salinity and low salinity populations of *Acartia tonsa*. These sets of geographically proximate sites potentially provide independent comparisons of the effects of seasonality across salinity gradients. Shown here are the collection conditions for these pairs of sites. Temperature was typically similar across the pairs within each collection, while salinity differences were fairly stable across collections.



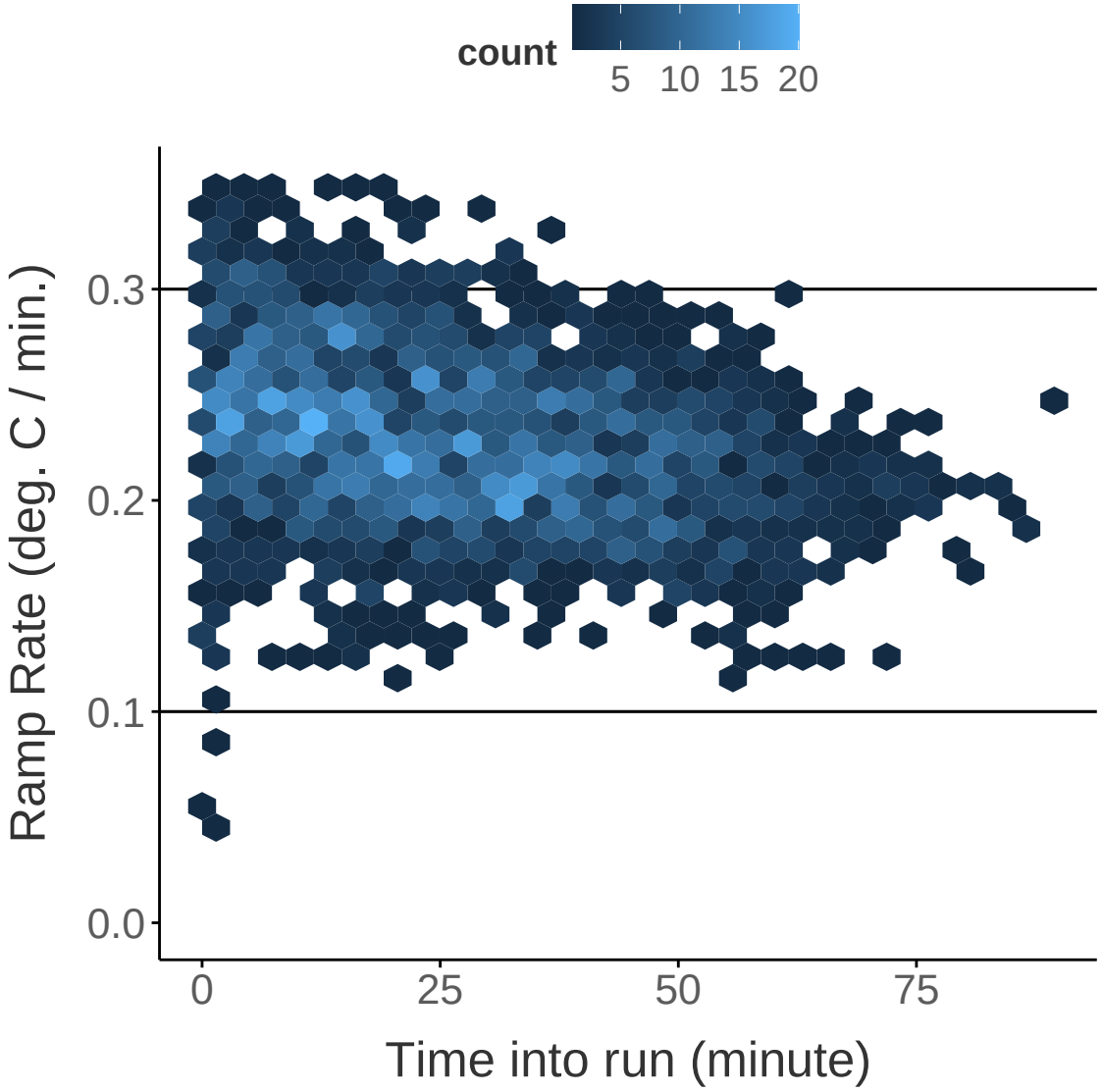
The three pairs of cross-salinity comparisons do show evidence for fine-scale trait divergence, although there was no consistent pattern in the direction or magnitude of differences. CTmax was similar across sites in the Southern and Central pairs. In the Northern pair, CTmax tended to be slightly lower for individuals from the low salinity site. Size was more variable between the paired sites. In the South, individuals from the low salinity site were consistently smaller than individuals from the high salinity site, despite experiencing similar temperatures. By contrast, in the Central pair individuals from the low salinity site tended to be slightly

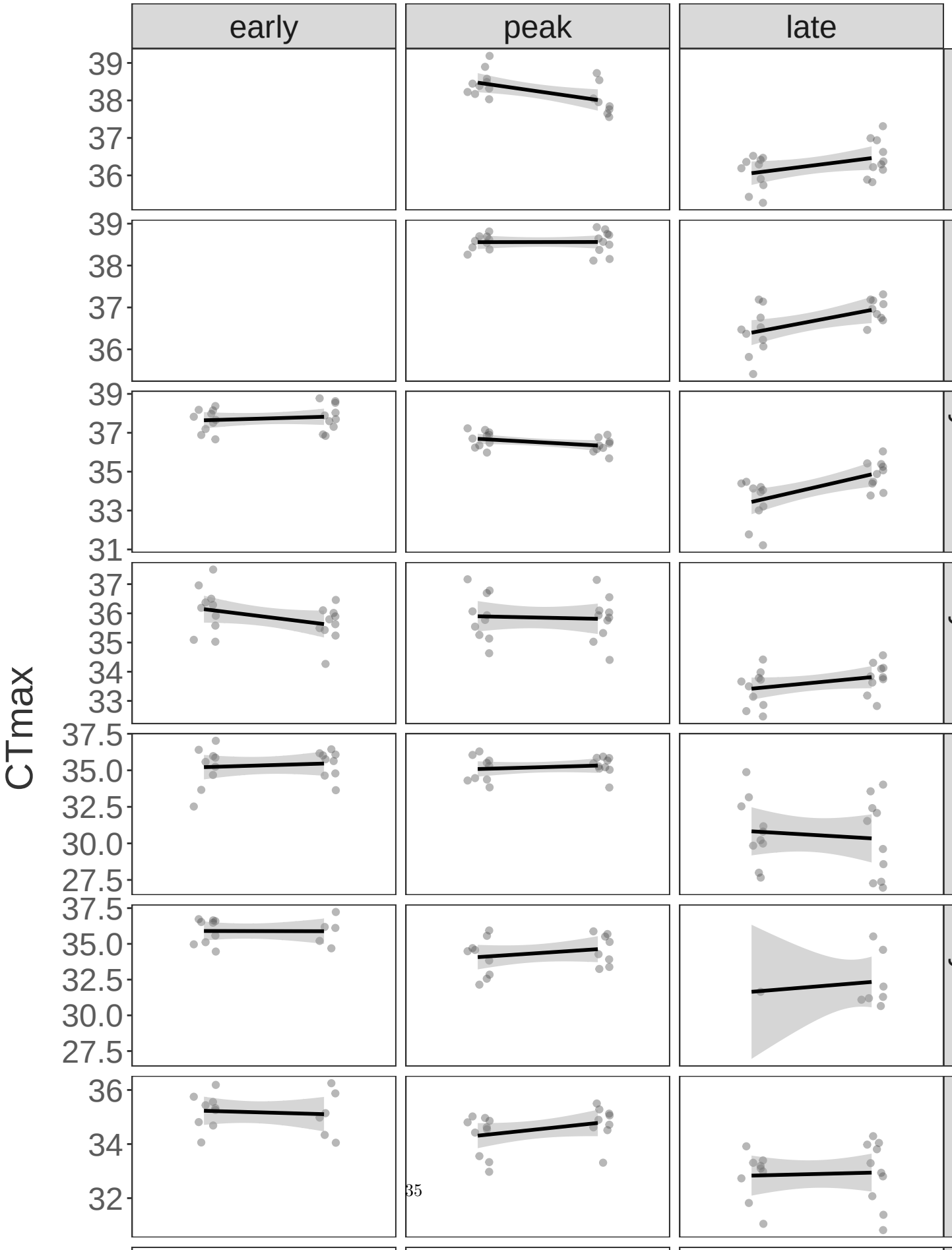
larger than those from the high salinity site, although the magnitude of this difference this varied seasonally. In the Northern pair, individuals from the low salinity site were slightly larger than individuals from the high salinity site during the early and peak season collections, but there was an abrupt reversal of this pattern during the late collection, when individuals from the high salinity site were much larger than individuals from the low salinity site.

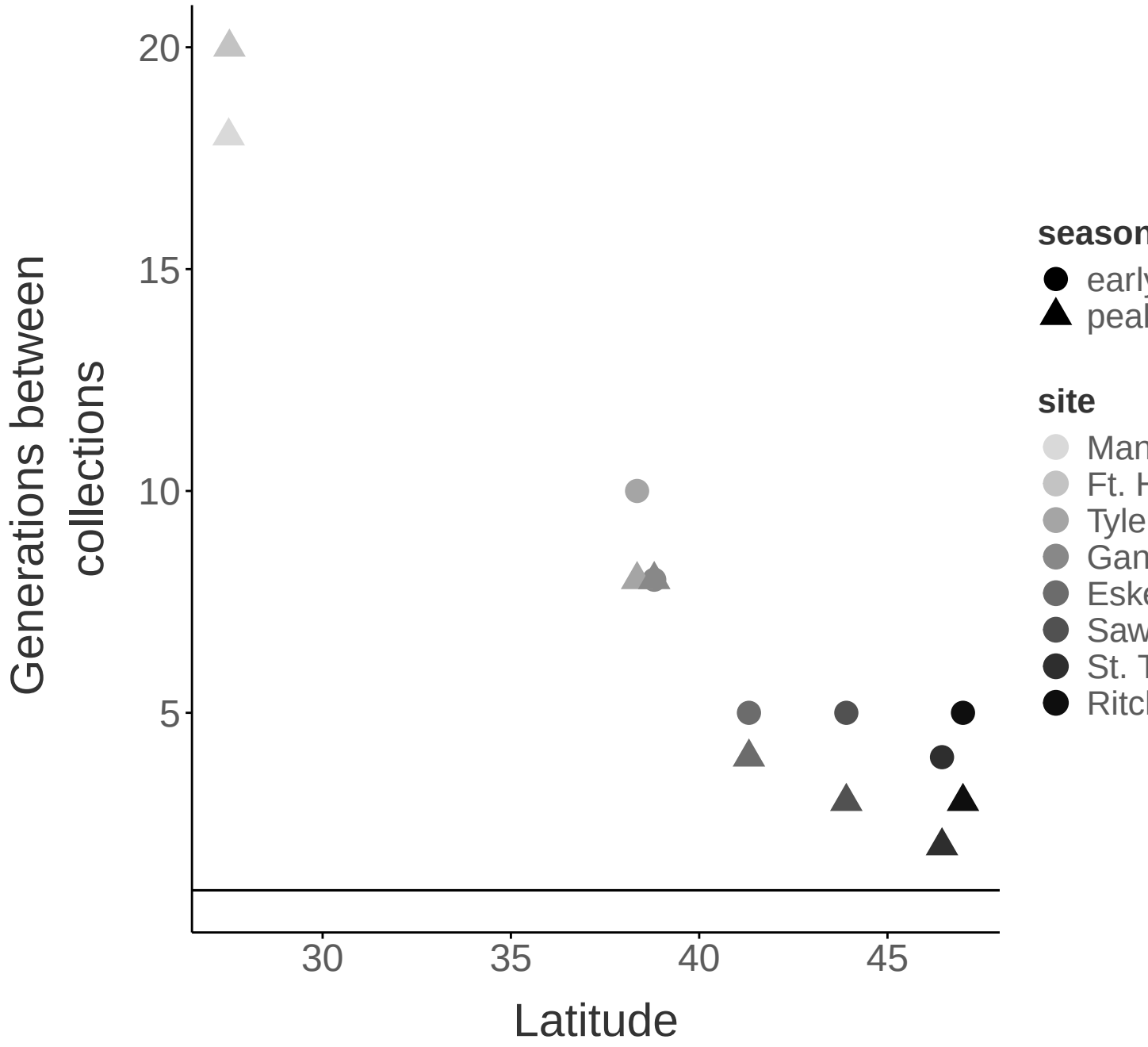
We will return to potential mechanisms behind divergence between these geographically proximate but environmentally separated sites later in this report.

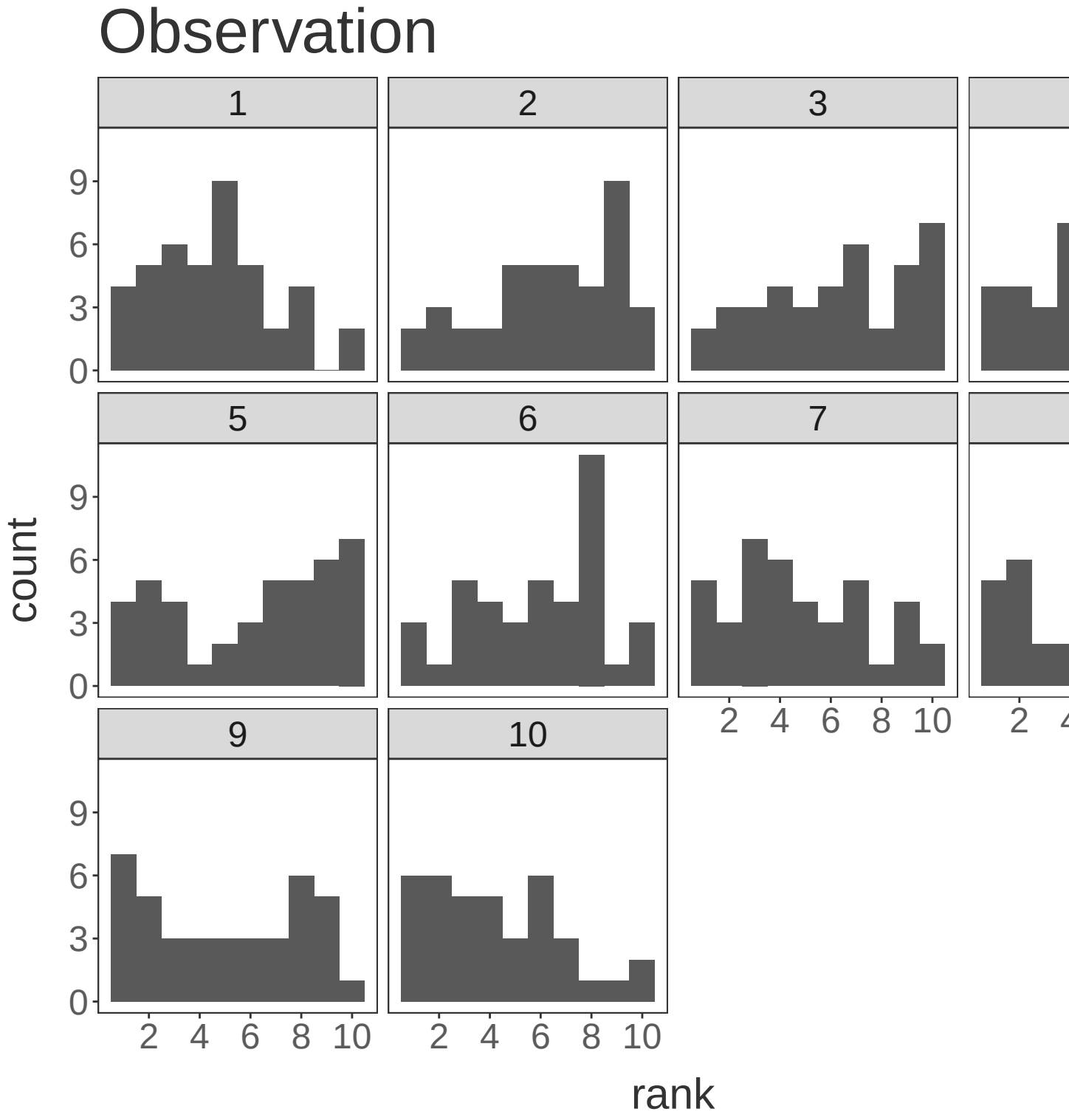


Misc. Details




early
peak
late
CTmax
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Generations between collections
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Latitude
season
early
peak
site
Man
Ft. H
Tyle
Gan
Eske
Saw
St. T
Ritch


Observation
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